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MISO DPP April 2018 East (ATC) Phase 3

System Impact Study Report

**J986, J1000, J1002, J1003, J1042, J1101, J1153, J1154, J1171, J1183,
and J1188**

Prepared for MISO

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1.0 EXECUTIVE SUMMARY

Eleven (11) resource projects have requested to interconnect to the Midcontinent Independent System Operator (MISO) transmission network in the East (ATC) Area and are included in the Definitive Planning Phase 2018 April Phase 3 study (Apr 18 DPP Phase 3). All Generating Facilities, except one, have requested both Energy Resource Interconnection Service (ERIS) and Network Resource Interconnection Service (NRIS). J1183 has requested only ERIS.

This report presents the study results of a System Impact Study (SIS) performed to evaluate the interconnection of the generator interconnection requests in the Apr 18 DPP Phase 3 study. The study was performed under the direction of MISO. The results for 2023 scenario are summarized below.

1.1 Project List

The Apr 18 DPP Phase 3 has eleven (11) interconnection requests with a combined nameplate rating of 925.11 MW. The Apr 18 DPP Phase 3 generator interconnection requests are listed in Table 1.1-1. The one-line diagrams of the interconnection facilities are shown in Appendix C.



Table 1.1-1 – Generating Facilities in DPP April 2018 Phase 3 East (ATC) Area

MISO Queue #	Service Type	Control Area	County, State	Point of Interconnection	Fuel Type	Inverter Model	Main Step-Up Transformer	Requested MW	Dispatch (MW) ¹			
									Batteries Discharging		Batteries Charging	
									23SH	23SUM	23SH	23SUM
J986	NRIS	ALTE	Wood, WI	Port Edwards - Sand Lake 138-kV	Solar	50 x 3.18 MVA TMEIC Ninja 840 PVU-L0840GR Total Rated MVA = 159	One 34.5/138 kV 105/140/175 MVA Z=5.75%	149.76	74.88	149.76	74.88	149.76
J1000	NRIS	ALTE	Grant, WI	Nelson Dewey - Lancaster 138-kV	Solar	17 x 3.5 MVA Power Electronics HEC-US V1500 FS3000CU15 Total Rated MVA = 58.3	One 34.5/138 kV 33/45/55 MVA Z=9.5%	50	25	50	25	50
J1002	NRIS	ALTE	Waushara, WI	Wautoma 138-kV	Solar	33 x 3.5 MVA Power Electronics Freesun HEC-US V1500 FS3000CU15 Total Rated MVA = 115.5	One 34.5/138 kV 66/88/110 MVA Z=9.5%	99	49.5	99	49.5	99
J1003	NRIS	ALTE	Dodge, WI	North Beaver Dam 69-kV	Solar	17 x 3.5 MVA Power Electronics Freesun HEC-US V1500 FS3000CU15 Total Rated MVA = 58.3	One 34.5/138 kV 33/55 MVA Z=8%	50	25	50	25	50
J1042	NRIS	ALTE	Walworth, WI	North Lake Geneva 138-kV	Solar	50 x 4.05 MVA TMEIC Ninja 840 Total Rated MVA = 203	One 138/34.5/13.8 kV 135/180/225 MVA Z=8.5%	180	90	180	90	180
J1101	NRIS	WPS	Manitowoc, WI	Kewaunee 138-kV	Battery	17 x 1.29 MVA Power Electronics Fremaq FP1290 Total Rated MVA = 21.93	One 34.5/138 kV 15/20/25 MVA Z=8%	20	20	20	-20	-20
J1153	NRIS	WEC	Sheboygan, WI	Holland 138-kV	Solar	50 x 3.15 MVA Sungrow SG3150U Total Rated MVA = 157.5	One 34.5/138 kV 96/128/160 MVA Z = 10%	150	75	150	75	150
J1154	NRIS	WEC	Jefferson, WI	Jefferson 138-kV	Solar	25 x 3.15 MVA Sungrow SG3150U Total Rated MVA = 78.75	One 34.5/138 kV 66/88/111 MVA Z = 10%	75	37.5	75	37.5	75
J1171	NRIS	WEC	Dodge, WI	Butternut 138-kV	Solar	36 x 3 MVA Power Electronics HEC-US V1500 FS3001CU15 Total Rated MVA = 108	One 34.5/138 kV 100/120 MVA Z=6%	100	50	100	50	100
J1183	ERIS	MIUP	Delta, MI	Garden Corners 138-kV	Solar	27 x 0.063 MVA CPS/Chint Power 50KTL Total Rated MVA = 1.69	Existing	1.35	0.675	1.35	0.675	1.35
J1188	NRIS	ALTE	Rock, WI	Sheepskin 69-kV	Solar	18 x 3 MVA Power Electronics HEC-US V1500 FS3001CU15 Total Rated MVA = 54	One 34.5/69 kV 33/44/55 MVA Z=9.0%	50	25	50	25	50

¹ Per MISO BPM 015-r21 the following dispatch assumptions are applied for each Fuel Type.

- a. Combined Cycle (CC) is dispatched to 50% of the Requested MW in the shoulder models, 100% in the summer peak models.
- b. Solar is dispatched to 50% of the Requested MW in the shoulder models, 100% in the summer peak models.
- c. Wind is dispatched to 100% of the Requested MW in the shoulder models, 15.6% in the summer peak models.
- d. Battery is dispatched, both in charging and discharging models, to 100% of the Requested MW in all study models.

1.2 Generating Facility Requirements

1.2.1 Voltage Schedule Requirement

ATC (Transmission Owner) requires all generators in its territory to maintain a voltage schedule at the Point of Interconnection (POI). The standard voltage schedule is 1.02 per unit as measured at the POI. This schedule may be changed by the Transmission Operator for specific power plants or specific conditions.

1.2.2 Power Factor Range Requirement

FERC Order 827 and ATC Criteria require all newly interconnecting generators interconnecting to ATC-owned Facilities to provide a power factor range for synchronous and non-synchronous (e.g. wind turbines, solar) generation of 0.95 leading (when a Generating Facility is consuming reactive power from the Transmission System) to 0.95 lagging (when a Generating Facility is supplying reactive power to the Transmission System). The Generating Facility must be capable of maintaining ATC's standard power factor range at all power output levels by providing dynamic reactive power at the following locations:

- A. The POI for all synchronous generators
- B. The high-side of the generator substation for all non-synchronous generators

For synchronous machines, the interconnection studies will account for the net effect of all energy production devices and losses on the Customer's side of the POI. For non-synchronous machines, the interconnection studies will account for the net effect of all energy production devices and losses on the Customer's side of the generator substation. Dynamic reactive power provided by non-synchronous generators must meet the following requirement from FERC order 827 Item 35:

"Non-synchronous generators may meet the dynamic reactive power requirement by utilizing a combination of the inherent dynamic reactive power capability of the inverter, dynamic reactive power devices (e.g., Static VAR Compensators), and static reactive power devices (e.g., capacitors) to make up for losses."

Therefore, static reactive power sources can only be used to make up for losses between the terminal of the machines and the high side of the generator substation for non-synchronous machines. All other reactive power to meet the power factor requirement must be provided by dynamic sources. Static sources can be switched on or off in the range of seconds and provide reactive power in large discrete blocks. Capacitor Banks are considered static sources of reactive power. Dynamic sources can provide variable amounts of reactive power in a few milliseconds. Static Var Compensators (SVCs), Static Synchronous Compensators (STATCOMs), Flexible AC Transmission Systems (FACTS), inverters and synchronous condensers are all considered dynamic sources of reactive power.

Non-synchronous generation projects in the Apr 18 DPP Phase 3 study group are required to provide dynamic reactive power within the range of 0.95 leading to 0.95 lagging at the high-side of the generator substation.

The generation requests shown in Table 1.2.2-1 did not meet the dynamic reactive power requirements per FERC Order 827 and ATC Criteria at the time of the model completion and are required to provide additional dynamic reactive power sources. All other requests in this queue met FERC Order 827 and ATC Criteria for dynamic reactive power requirements.

Table 1.2.2-1 – Additional Dynamic Mvar to meet ATC Dynamic Inductive Power Factor Requirement and FERC Order 827 Power Factor Requirements

MISO Queue #	Type	Additional Dynamic Reactive Compensation (Mvar)
J1188	Asynchronous	-0.95

The generation requests shown in Table 1.2.2-2 did not meet the static reactive power requirements per ATC Criteria at the time of the model completion and are required to provide additional static reactive power sources. All other requests in this queue met FERC Order 827 and ATC Criteria for static reactive power requirements.

Table 1.2.2-2 – Additional Static Mvar to meet ATC Capacitive Power Factor and FERC Order 827 Power Factor Requirements

MISO Queue #	Type	Additional Static Shunt Compensation ¹ (Mvar)
J1042	Asynchronous	5.0
J1101	Asynchronous	0.7
J1153	Asynchronous	34.1
J1154	Asynchronous	13.7
J1171	Asynchronous	13.9
J1188	Asynchronous	10.13

¹ Additional compensation is required to meet the criteria at the POI Bus for synchronous Generating Facilities or the high-side of the generator substation for asynchronous Generating Facilities.

No generation requests need an assessment of power factor at low real power output levels.

1.2.3 Island Detection and Operation

In circumstances where the Generating Facility has no governor controls and the transmission system design could result in an islanding condition for the outage of two transmission elements, ATC will develop operating solutions requiring generation curtailment when the next contingency could result in an island condition. ATC continues to work with developers and evaluate technical

solutions (e.g. requiring customer to implement additional protection systems) that could be considered as alternatives to implementing the operating solutions.

This would apply to the following Generating Facilities from this DPP cycle that lack adequate governor controls to safely and reliably sustain an island with load.

- J1000
- J1153
- J1171

1.3 Total Network Upgrades for All Projects

The cost allocation of Network Upgrades for the study group reflects responsibilities for mitigating system impacts based on Interconnection Customer-elected level of Energy Resource Interconnection Service and Network Resource Interconnection Service as of the SIS report date. The total cost of Network Upgrades required for each generator interconnection request is listed in Table 1.3-1. The costs for Network Upgrades except for Interconnection Facilities costs are planning level estimates and subject to be revised in the facility studies. All Interconnection Facility Project Diagrams are documented in Appendix C and all Network Upgrade Project Diagrams are documented in Appendix D (No project diagrams are developed for line upgrades).

Table 1.3-1 – Total Cost of Network Upgrades for April 18 DPP Phase 3 Generator Interconnection Requests

MISO Queue #	Requested MW	ERIS Network Upgrades (\$)				NRIS Network Upgrades (\$)	Interconnection Facilities (\$)		Shared Network Upgrade (\$)	Total Cost of Network Upgrades (Exclude TOIF & Affected Systems) (\$)	M2 Received (\$) ¹	M3 Received (\$) ²	M4 Received (\$) ³
		Steady - State Thermal & Voltage	Transient Stability	Short Circuit	Affected System ⁸		TO Network Upgrades	TO-Owned Direct Assigned (TOIF)					
[a]	[b]	[c]	[d]	[e]	[f]	[g]	[h]	[i]	[j]	[k]=[c]+[d]+[e]+[g]+[h]	[l] = \$4000 x [b]	[m]= (10%of [k])-[l]	[n]= (20%of [k])-[l] –[m]
J986	149.76	5,093,272	0	1,020,668	0	0	6,740,635	258,407	0	12,854,575	599,040	776,690	1,180,439
J1000	50	0	0	0	0	0	6,441,458	252,140	0	6,441,458	200,000	0	1,088,292
J1002	99	1,575,471	0	737,572	0	0	2,618,981	627,686	0	4,932,024	396,000	0	577,015
J1003	50	0	0	221,760	0	0	366,697	664,689	0	588,457	200,000	0	437,691
J1042	180	0	0	220,000	0	3,720,312	750,098	627,259	0	4,690,410	800,000	258,992	1,944,069
J1101	20	0	0	0	0	0	0	0	0	0	80,000	0	0
J1153	150	0	0	660,000	0	0	4,106,408 ⁹	651,073	0	4,766,408	600,000	0	221,282
J1154	75	1,860,217	0	0	0	0	0	0	0	1,860,217	800,000	0	402,638
J1171	100	0	0	220,000	0	0	1,521,698 ⁹	819,868	0	1,741,698	400,000	0	965,375
J1183	1.35	0	0	0	0	0	0	0	0	0	5,400	0	0
J1188	50	0	0	220,000	320,211 ⁷	243,282	1,362,993 ¹⁰	267,141 ¹⁰	0	1,826,275	200,000	135,899	921,327
Total (\$)	-	8,528,960	0	3,300,000	320,211	3,963,594	23,908,968	4,168,263	0	39,701,522	3,780,440	1,063,357	3,375,581

¹ M2: Milestone Payment dollars received by MISO.

² M3 = (10% of NU)-M2.

³ M4= (20% of NU)-M2-M3.

⁴ ATC projects are estimated in ISD dollars with a 2.5% annual escalation rate and include a 10% contingency.

⁵ Transmission Owner shall also collect from Interconnection Customer a tax gross-up amount on the payments made to Transmission Owner using the Transmission Owner rate in effect at the time the payment is received from Interconnection Customer. The current Transmission Owner tax gross-up rate is 12.848%.

⁶ TOIF: Non-reimbursable Transmission Owner Interconnection Facilities to which tax gross-up amount must also be applied.

⁷ Rock Energy Coop and Alliant Energy Affected System upgrade cost based on the scope of work defined in Phase 2 facility study report, REV 0 dated 2/21/20.

⁸ Any potential Affected System upgrades and associated costs would need to be determined by Affected System Facility Owner and are not included in this table.

⁹ Additional protection scopes for J1153 and J1171 interconnection facilities were identified after the April-18 Phase 2 Facility Study and these costs will be updated in addendums to Phase 2 Facility Study Reports.

¹⁰ These costs were based on the scope of work defined in Phase 2 facility study report, REV 0 dated 2/21/20. They will be updated in Phase 2 facility study report REV 1 due to J1188 generator lead line reroute.

1.4 Stability Operating Restrictions Summary

For NERC Category P6 stability constraints, operating restrictions may be required under prior outage conditions to avoid instability caused by the next event. A summary of all P6 stability operating restrictions identified for study generators in the Apr 18 DPP WI Phase 3 are provided in Table 1.4-1. Maximum allowed real power outputs to mitigate P6 stability constraints under prior outage conditions were identified for each generating facility. These constraints are discussed in further detail in Sections 6.3.3.

Table 1.4-1 – NERC Category P6 Operating Restriction Analysis Result Summary

MISO Queue #	Max MW	NERC Category P6 (N-1-1) Event		Stability Max MW Allowed
		1 st Contingency	2 nd Contingency	
J986	150	[REDACTED]	[REDACTED]	100
J986 & J1002	249	[REDACTED]	[REDACTED]	200

1.5 In-Service Dates and Cost Estimates

ATC understands that the estimated in-service date may not align with the Interconnection Customer's Synchronization Date; however, negotiated and executed agreements, such as an Engineering and Procurement Agreement, can be used prior to the GIA execution date to expedite Network Upgrades. In absence of any special arrangement, typical times to develop a new Interconnection Facility is about 24-36 months after the GIA is executed, assuming no delays due to Interconnection Customer's permits, state processes, land acquisitions, deliverables (such as a finish graded substation site, etc.) It also assumes that system outages required to construct facilities can be obtained timely. The cost estimates for Interconnection Facilities are based on the in-service date provided in the Interconnection Customer's application data. Therefore, any change in in-service date will have impact on the cost estimates. The requested dates for Interconnection Facility in-service, synchronization, and commercial operation are summarized in Table 1.5-1.

Table 1.5-1 – Requested Interconnection Facilities In-Service Dates, Synchronization Dates and Commercial Operation Dates

MISO Queue #	Requested Interconnection Facility In-service Date	Requested Synchronization Date	Requested Commercial Operation Date
J986	September 1, 2020	September 15, 2020	December 1, 2020
J1000	October 1, 2021	October 15, 2021	December 1, 2021
J1002	October 1, 2021	October 15, 2021	December 1, 2021
J1003	October 1, 2021	October 15, 2021	December 1, 2021

J1042	September 1, 2023	October 1, 2023	December 1, 2023
J1101	September 1, 2020	October 1, 2020	December 1, 2020
J1153	April 1, 2021	April 15, 2021	June 30, 2021
J1154	April 1, 2021	April 15, 2021	June 30, 2021
J1171	August 1, 2020	October 1, 2020	December 1, 2020
J1183	January 31, 2019	January 31, 2019	January 31, 2019
J1188	August 31, 2021	September 30, 2021	December 1, 2021

1.6 MTEP Projects

If a MISO Transmission Expansion Plan (MTEP) project(s) resolves the constraint, and that project(s) is approved by the MISO Board within (1) calendar year of the GIA execution or execution of an amendment thereof, then the Interconnection Customer will not be responsible for transmission upgrade(s) that would resolve the constraint. If that MTEP project(s) is not approved within one (1) calendar year of the GIA execution or execution of an amendment thereof, the Interconnection Customer will be responsible for those transmission upgrade(s).

MISO performed analysis and monitored upgrades that are not yet in service to identify upgrades which the study cycle projects are conditional upon in order to meet their requested service levels. Results are documented in Appendix M.

1.7 Further Study

The next step in the MISO Generator Interconnection Procedures is to perform Network Upgrade Facility Studies. Those Facilities Studies will specify in more detail the time and cost of the equipment, engineering, procurement, and construction of the Network Upgrades identified in this report.

1.8 Compliance Summary

This study report partially meets NERC TPL-001-4 standard, FAC-002-2 standard, and ATC Planning Criteria. In ATC's annual Ten-Year Assessment (TYA) and MISO annual MTEP studies, additional compliance related studies will be performed for the generator interconnection requests with signed GIAs. Appendix J describes in detail the NERC standards and ATC Planning Criteria requirements met by this SIS report.

2.0 STEADY-STATE ANALYSIS

Steady-state analysis was performed to identify thermal and voltage upgrades required to interconnect the generator interconnection requests in the Apr 18 DPP Phase 3 to the transmission system. Detailed study assumptions, criteria, and methodologies are documented in Appendix A.

2.1 Model Development

2.1.1 Study Cases

Using the final DPP Apr 18 Phase 2 study cases as the starting point, four Phase 3 study cases for the steady-state thermal and voltage analysis were developed: 2023 summer peak batteries discharging, 2023 summer peak batteries charging, 2023 shoulder batteries discharging, and 2023 shoulder batteries charging. Major updates include:

- Removed J807, J819 and J821 due to Post-GIA withdrawals.
- Removed the following Network Upgrades based on 2017 August WI cycle restudy results.
 - o North Monroe - Bass Creek 138 kV, uprate
 - o Darlington - North Monroe uprate
 - o South Monroe - Browntown rebuild
 - o Rock River SS terminal upgrade
- Removed generation requests in ATC that had dropped out of the MISO queue since the final Phase 2 study models were built. Removed any related Network Upgrades.
- Incorporated study generator modeling changes. J1042 reduced its MW request from 200 MW to 180 MW. J1153 and J1154 changed inverters.

The Cardinal – Hickory Creek project was included in all study models, even though its in-service date is 12/31/2023. This is because it was defined as a required Network Upgrade in the DPP 2017 February Wisconsin Area Phase 1 SIS. The cases were reviewed by MISO, ATC and the Interconnection Customers. Based on this review, the cases were further modified to account for additional model updates.

The prior queued generator interconnection requests in the ATC system that are included in the study cases are listed in Table 2.1.1-1. Associated Network Upgrades were also included based on their expected in-service date.

**Table 2.1.1-1 – Prior Queued Generator Interconnection Requests
Not Yet In-Service, as of Apr 18 DPP Phase 1**

MISO Queue #	Type	Control Area	Requested MW
J390	CC	ALTE	702
J505	Solar	WPS	99
J584	Wind	ALTE	60
J732	CC	WPS	561.5
J760	CC	ALTE	30
J818	Solar	WEC	149
J831	CC	WEC	48
J849	Solar	MIUP	125

J850	Solar	ALTE	250
J855	Wind	ALTE	100
J864	Solar	ALTE	49.98
J870	Solar	ALTE	200
J871	Solar	ALTE	100
J878	Solar	WEC	200
J886	Solar	WPS	150
J928	Wind	MIUP	79.995
J947	Solar	ALTE	200

Public information related to the MISO Generator Interconnection Request queue can be found at:

https://www.misoenergy.org/planning/generator-interconnection/GI_Queue/

The four Phase 3 study cases (summer peak batteries discharging, summer peak batteries charging, shoulder batteries discharging, and shoulder batteries charging) dispatched generation within MISO according to section 6.1.1.1.2, Study Case Development, in the MISO BPM-015-r21.

All excess generation from this methodology is dispatched against all units in MISO Classic proportionally, excluding the units in the current DPP cycle. Scheduled firm transfers are ignored in this dispatch methodology.

2.1.2 Benchmark Cases

Using the final DPP Apr 18 Phase 2 benchmark cases as the starting point, two benchmark cases were used to benchmark system performance without the Apr 18 DPP Phase 3 generating facilities and were created by taking the Apr 18 DPP Phase 3 Generating Facilities offline from the two discharging study cases. The MISO Classic was used for power balance, where generation was scaled in proportion to $P_{max} - P_{gen}$. The major updates listed in Section 2.1.1 were also incorporated in the two Benchmark cases.

2.2 Reactive Power Requirements (FERC Order 827)

All synchronous and non-synchronous generation in this queue were evaluated to determine if the requests meet FERC Order 827 and ATC Planning Criteria. Refer to PLG-METH-0005 in Appendix B for details on the methodology used to determine power factor compliance. All of the reactive resources modeled in the assessment are summarized in Table 2.2-1.

Table 2.2-1 – Reactive Resources Modeled in Generator Interconnection Power Factor Analysis

MISO Queue #	Machines				Dynamic Reactive Devices			Static Reactive Devices		
	Description	Real Power (MW)	Capacitive Reactive Power (Mvar)	Inductive Reactive Power (Mvar)	Description	Capacitive Reactive Power (Mvar)	Inductive Reactive Power (Mvar)	Description	Capacitive Reactive Power (Mvar)	Inductive Reactive Power (Mvar)
J986	Solar	149.937	52.91	-52.91	None	N/A	N/A	6 * 4.2 Mvar Capacitor Banks	25.2	0
J1000	Solar	50	29.668	-29.668	None	N/A	N/A	None	0	0
J1002	Solar	99	59	-59	None	N/A	N/A	None	0	0
J1003	Solar	50	29.668	-29.668	None	N/A	N/A	None	0	0
J1042	Solar	180	60.0	-60.0	None	N/A	N/A	2 x 14 Mvar Capacitor Banks	28.0	0
J1101	Battery	20	6.574	-6.574	None	N/A	N/A	3 Mvar Capacitor Bank	3	0
J1153	Solar	150	48.02	-48.02	None	N/A	N/A	None	0	0
J1154	Solar	75	24.01	-24.01	None	N/A	N/A	None	0	0
J1171	Solar	102.58	33.92	-31.82	None	N/A	N/A	None	0	0
J1183	Solar and Wind ¹	103.015	39.0862	-39.0862	None	N/A	N/A	10.5 Mvar, 2 x 4.8 Mvar Capacitor Banks	20.1	0
J1188	Solar	51.29	16.96	-15.91	None	N/A	N/A	None	0	0

¹ Existing wind generations behind the J1183 POI are also considered for the analysis.

The dynamic capacitive power factor requirement analysis showed all requests meeting ATC Planning Criteria and FERC Order 827 requirements. The results are summarized in Table 2.2-2.

Table 2.2-2 – Assessment of Dynamic Capacitive Power Factor Requirement

MISO Queue #	Bus #	Capability at Machine Terminal Bus		Additional Dynamic Capacitive Reactive Power ¹ (Mvar)	Dynamic Power Factor Provided	Meets Requirement ² ?	Additional Requirement ³ (Mvar)
		Real Power (MW)	Capacitive Reactive Power (Mvar)				
J986	89860	149.9	52.9	0.0	0.94	Yes	0.0
J1000	40000	50.0	29.67	0.0	0.86	Yes	0.0
J1002	40020	99.0	58.743	0.0	0.86	Yes	0.0
J1003	40030	50.00	29.7	0.0	0.86	Yes	0.0
J1042	40420	200.0	60.0	0.0	0.95	Yes	0.0
J1101	41010	20.000	6.574	0.0	0.95	Yes	0.0
J1153	41530	150.0	48.02	0.0	0.95	Yes	0.0
J1154	40540	75	24.01	0.0	0.95	Yes	0.0
J1171	40710	102.58	33.92	0.0	0.95	Yes	0.0
J1183	41830 693697 693704 693429	103.015	39.0862	0.0	0.93	Yes	0.0
J1188	41880	51.29	16.96	0.0	0.95	Yes	0.0

¹ Dynamic capacitive reactive power provided by Interconnection Customer owned equipment in addition to the machine.

² ATC requires a 0.95 ATC Capacitive Dynamic Power Factor.

³ Additional dynamic reactive power required to meet ATC Capacitive Dynamic Power Factor.

The static capacitive power factor requirement analysis showed 6 of the 11 requests do not meet the ATC Criteria or FERC Order 827 requirements. The results are summarized in Table 2.2-3.

Table 2.2-3 – Assessment of Static Capacitive Power Factor Requirement

MISO Queue #	Point of Measurement	Capability at Point of Measurement ¹		Power Factor	Meets Requirement?	Additional Requirement (Mvar)
		Real Power (MW)	Reactive Power (Mvar)			
J986	89863	147.1	51.6	0.94	Yes	0.0
J1000	40003	49.22	19.9	0.93	Yes	0.0
J1002	40023	97	37	0.93	Yes	0.0
J1003	40033	49	20.2	0.92	Yes	0.0
J1042	40420	177.5	53.3	0.96	No	5.0
J1101	88604	19.8	5.8	0.96	No	0.7
J1153	41533	147.8	14.5	1.00	No	34.1
J1154	41543	73.9	10.6	0.99	No	13.7
J1171	41713	100	19	0.98	No	13.9
J1183	693702	100.5	36.9	0.94	Yes	0.0
J1188	41883	50.64	6.51	0.99	No	10.13

¹ Point of Measurement is the POI Bus for synchronous machines and high side of generator substation for asynchronous machines.

The dynamic inductive power factor requirement analysis showed all but two of the requests meet ATC Criteria and FERC Order 827 requirements. J1188 did not meet the ATC Criteria and FERC Order 827 requirements and must provide additional dynamic resources. The results are summarized in Table 2.2-4.

Table 2.2-4 – Assessment of Dynamic Inductive Power Factor Requirement

MISO Queue #	Bus #	Capability at Machine Terminal		Additional Dynamic Inductive Reactive Power ¹ (Mvar)	Dynamic Power Factor Provided	Meets Requirement ² ?	Additional Requirement ³ (Mvar)
		Real Power (MW)	Inductive Reactive Power (Mvar)				
J986	89860	149.9	-52.9	0.0	0.94	Yes	0.0
J1000	40000	50.0	-29.67	0.0	0.86	Yes	0.0
J1002	40020	99.0	-58.7431	0.0	0.86	Yes	0.0
J1003	40030	50.00	-29.7	0.0	0.86	Yes	0.0
J1042	40420	180	-60	0.0	0.95	Yes	0.0
J1101	41010	20.000	-6.574	0.0	0.95	Yes	0.0
J1153	41530	150.0	-48.02	0.0	0.95	Yes	0.0
J1154	40540	75	-24.01	0.0	0.95	Yes	0.0
J1171	40710	102.58	-46.1	0.0	0.91	Yes	0.0
J1183	41830 693697 693704 693429	103.015	-39.0862	0.0	0.93	Yes	0.0
J1188	41880	51.29	-15.91	0.0	0.96	No	-0.95

¹ Dynamic inductive reactive power provided by Interconnection Customer owned equipment in addition to the machine.

² ATC requires a 0.95 ATC Inductive Dynamic Power Factor.

³ Additional dynamic reactive power required to meet ATC Inductive Dynamic Power Factor.

The static inductive power factor requirement analysis showed all requests meeting ATC Criteria and FERC Order 827 requirements. The results are summarized in Table 2.2-5.

Table 2.2-5 – Assessment of Static Inductive Power Factor Requirement

MISO Queue #	POI Bus (synchronous) or HV Bus (asynchronous)	MW at POI (synchronous) or HV Bus (asynchronous)	Mvar at POI (synchronous) or HV Bus (asynchronous)	POI (synchronous) or HV Bus (asynchronous) Power Factor	Meets 0.95 ATC Inductive Power Factor Requirement at POI Bus (synchronous) or HV Bus (asynchronous)?
J986	89863	147.1	-80.3	0.88	Yes
J1000	40003	49.0	-42.94	0.75	Yes
J1002	40023	96.4	-88.4	0.74	Yes
J1003	40033	48.7	-42.6	0.75	Yes
J1042	40420	177.4	-94.5	0.88	Yes
J1101	886004	19.7	-10.2	0.89	Yes
J1153	41533	147.6	-84.1	0.87	Yes
J1154	41543	73.9	-37.6	0.89	Yes
J1171	41713	100.0	-46.1	0.91	Yes
J1183	693702	100.3	-61.7	0.85	Yes
J1188	41883	50.62	-26.7	0.88	Yes

The low output power factor requirement analysis showed all requests meet ATC Planning Criteria and FERC Order 827 requirements. The results are summarized in Table 2.2-6.

Table 2.2-6 – Assessment of Power Factor Requirements at Low Output Levels

MISO Queue #	P-Q Curve Type	Is Power Factor Evaluation Needed for Minimum Output Levels?	Additional Static Mvar required at POI Bus (synchronous) or HV Bus (asynchronous) to meet ATC Power Factor Requirement
J986	D-Shape	no	N/A
J1000	D-shape	no	N/A
J1002	D-shape	no	N/A
J1003	D-shape	no	N/A
J1042	D-shape	no	N/A
J1101	D-shape	no	N/A
J1153	D-shape	no	N/A
J1154	D-shape	no	N/A
J1171	D-shape	no	N/A
J1183	D-shape	no	N/A
J1188	D-shape	no	N/A

2.3 NERC TPL Contingency Analysis Results

The incremental impact of the proposed generator interconnection on transmission facilities was evaluated by comparing steady state power flows and voltages between benchmark cases (without Apr 18 DPP Phase 3 projects) and study cases (with Apr 18 DPP Phase 3 projects). Post-contingency cases were solved with transformer tap adjustment enabled, area interchange adjustment disabled, phase shifter adjustment enabled, and switched shunt adjustment enabled. NERC TPL Category P contingencies that were studied are described in Table A.2.1-1 in Appendix A.

2.3.1 2023 Summer Peak Batteries Discharging

The study identified the steady-state thermal and voltage constraints that qualified as MISO Injection Constraints in the 2023 Summer Peak Discharging study model under NERC Category P0-P7 Planning Events (except NERC Category P3 and P6). Detailed steady-state power flow results and Injection Constraint determination can be found in Appendix E. A summary of the 2023 Summer Peak Discharging MISO Injection Constraints that require Network Upgrades is presented in Table 2.3.1. No voltage constraints were identified in the steady-state analyses.

Table 2.3.1 – 2023 Summer Peak Batteries Discharging Steady-State Injection Constraints Requiring Network Upgrades

Study Case	Overloaded Facility	Facility Owner	Resulted by "No Load Loss Allowed" ¹ NERC TPL Planning Events	Responsible Generator(s)
2023 Summer, Batteries Discharging	Sand Lake 138/69 kV transformer	ATC	P12	J986
	Wautoma 138/69 kV transformer	ATC	Base Case P12 P13 P21	J986 J1002
	J986 POI – Port Edwards 138 kV line	ATC	P12 P13	J986 J1002

¹ The "No Load Loss Allowed" NERC TPL Planning Events refer to all the Planning Events in NERC TPL-001-4 Table 1 that the interruption of Firm Transmission Service and Non-Consequential Load Loss are not allowed.

² Per ATC's Planning Criteria, for Planning Events that allow Non-Consequential load loss, short duration exposure to potential post-contingency overload represents an acceptable risk assuming system adjustment within that short period of time can eliminate the exceedance and restore facility loading within Long-Term emergency ratings.

2.3.2 2023 Summer Peak Batteries Charging

The study identified the steady-state thermal and voltage constraints that qualified as MISO Injection Constraints in the 2023 Summer Peak Charging study model under NERC Category P0-P7 Planning Events (except NERC Category P3 and P6). Detailed steady-state power flow results and Injection Constraint determination can be found in Appendix E. A summary of the 2023 Summer Peak Charging MISO Injection Constraints that require Network Upgrades is presented in Table 2.3.2. No voltage constraints were identified in the steady-state analyses.

Table 2.3.2 – 2023 Summer Peak Batteries Charging Steady-State Injection Constraints Requiring Network Upgrades

Study Case	Overloaded Facility	Facility Owner	Resulted by "No Load Loss Allowed" ¹ NERC TPL Planning Events	Responsible Generator(s)
2023 Summer, Batteries Charging	Sand Lake 138/69 kV transformer	ATC	P12	J986
	Wautoma 138/69 kV transformer	ATC	Base Case P12 P13 P21	J986 J1002
	J986 POI – Port Edwards 138 kV line	ATC	P12 P13	J986 J1002

¹ The "No Load Loss Allowed" NERC TPL Planning Events refer to all the Planning Events in NERC TPL-001-4 Table 1 that the interruption of Firm Transmission Service and Non-Consequential Load Loss are not allowed.

² Per ATC's Planning Criteria, for Planning Events that allow Non-Consequential load loss, short duration exposure to potential post-contingency overload represents an acceptable risk assuming system adjustment within that short period of time can eliminate the exceedance and restore facility loading within Long-Term emergency ratings.

2.3.3 2023 Shoulder Batteries Discharging

The study identified the steady-state thermal and voltage constraints that qualified as MISO Injection Constraints in the 2023 Shoulder Discharging study model under NERC Category P0-P7 Planning Events (except NERC Category P3 and P6). Detailed steady-state power flow results and Injection Constraint determination can be found in Appendix E. A summary of the 2023 Shoulder Discharging MISO Injection Constraints that require Network Upgrades is presented in Table 2.3.3. No voltage constraints were identified in the steady state analyses.

Table 2.3.3 – 2023 Shoulder Batteries Discharging Steady-State Injection Constraints Requiring Network Upgrades

Study Case	Overloaded Facility	Facility Owner	Resulted by "No Load Loss Allowed" ¹ NERC TPL Planning Events	Responsible Generator(s)
2023 Shoulder, Batteries Discharging	Saratoga 138/115 kV transformer T31	ATC	P23	J986
	Jefferson – Crawfish River 138 kV line	ATC	P12 P23	J1154
	Crawfish River – Concord 138 kV line	ATC	P12 P23	J1154

¹ The "No Load Loss Allowed" NERC TPL Planning Events refer to all the Planning Events in NERC TPL-001-4 Table 1 that the interruption of Firm Transmission Service and Non-Consequential Load Loss are not allowed.

² Per ATC's Planning Criteria, for Planning Events that allow Non-Consequential load loss, short duration exposure to potential post-contingency overload represents an acceptable risk assuming system adjustment within that short period of time can eliminate the exceedance and restore facility loading within Long-Term emergency ratings.

2.3.4 2023 Shoulder Batteries Charging

The study identified the steady-state thermal and voltage constraints that qualified as MISO Injection Constraints in the 2023 Shoulder Charging study model under NERC Category P0-P7 Planning Events (except NERC Category P3 and P6). Detailed steady-state power flow results and Injection Constraint determination can be found in Appendix E. A summary of the 2023 Shoulder Charging MISO Injection Constraints that require Network Upgrades is presented in Table 2.3.4. No voltage constraints were identified in the steady state analyses.

Table 2.3.4 – 2023 Shoulder Batteries Charging Steady-State Injection Constraints Requiring Network Upgrades

Study Case	Overloaded Facility	Facility Owner	Resulted by "No Load Loss Allowed" ¹ NERC TPL Planning Events	Responsible Generator(s)
2023 Shoulder, Batteries Charging	Saratoga 138/115 kV transformer T31	ATC	P23	J986
	Jefferson – Crawfish River 138 kV line	ATC	P12 P23	J1154
	Crawfish River – Concord 138 kV line	ATC	P12 P23	J1154

¹ The "No Load Loss Allowed" NERC TPL Planning Events refer to all the Planning Events in NERC TPL-001-4 Table 1 that the interruption of Firm Transmission Service and Non-Consequential Load Loss are not allowed.

² Per ATC's Planning Criteria, for Planning Events that allow Non-Consequential load loss, short duration exposure to potential post-contingency overload represents an acceptable risk assuming system adjustment within that short period of time can eliminate the exceedance and restore facility loading within Long-Term emergency ratings.

2.3.5 Network Upgrades Identified in ERIIS Analysis

Based on the steady-state analyses, the worst loading of each facility under "No Load Loss Allowed" NERC TPL Planning Events that meets MISO Injection Constraint criteria is shown in Table 2.3.5-1. Potential Network Upgrades are also included. Good faith Cost Estimates of the ERIIS thermal Network Upgrades identified in the steady-state analysis for the 2023 scenarios are listed in Table 2.3.5-2. Detailed cost allocations are provided in Section 9.

Table 2.3.5-1 – ERIS Network Upgrades Identified to Address MISO Steady-State Injection Constraints

Steady-State Injection Constraint	Responsible Generator(s)	Facility Owner	Study Case	Applicable Rating (MVA)	Worst Loading (%)	"No Load Loss Allowed" NERC TPL Planning Event	Network Upgrades
Sand Lake 138/69 kV transformer	J986	ATC	23SUM, Batteries Charging	60	104.9	[REDACTED]	Sand Lake 138/69 kV transformer, replacement
Wautoma 138/69 kV transformer	J986 J1002	ATC	23SUM, Batteries Charging	65	129.9	[REDACTED]	Wautoma 138/69 kV transformer, replacement
J986 POI – Port Edwards 138 kV line	J986 J1002	ATC	23SUM, Batteries Charging	153	102.5	[REDACTED]	J986 POI – Port Edwards 138 kV (X-11), uprate
Saratoga 138/115 kV transformer T31	J986	ATC	23SH, Batteries Discharging	239	103.2	[REDACTED]	MTEP PrjID 18017: Petenwell-Badger West-Saratoga 138kV (X-43), Rebuild
Jefferson – Crawfish River 138 kV line	J1154	ATC	23SH, Batteries Discharging	266	107.6	[REDACTED]	
Crawfish River – Concord 138 kV line	J1154	ATC	23SH, Batteries Charging	246	111.4	[REDACTED]	

Table 2.3.5-2 – ERIS Network Upgrades and Cost Estimates

Steady-State Injection Constraint	Facility Owner	Network Upgrade	Cost (\$)^{1,2}
Sand Lake 138/69 kV transformer	ATC	Sand Lake 138/69 kV transformer, replacement	3,105,696
Wautoma 138/69 kV transformer	ATC	Wautoma 138/69 kV transformer, replacement	2,978,370
J986 POI – Port Edwards 138 kV line	ATC	J986 POI – Port Edwards 138 kV (X-11), uprate	584,677
Saratoga 138/115 kV transformer T31	ATC	MTEP PrjID 18017: Petenwell-Badger West-Saratoga 138kV (X-43), Rebuild	0
Jefferson – Crawfish River 138 kV line	ATC	Jefferson – Crawfish River 138 kV (6654), uprate	957,298
Crawfish River – Concord 138 kV line	ATC	Crawfish River – Concord 138 kV (9043), uprate	902,919

¹ All Network Upgrades were estimated on the generator ISD dollars.

² ATC Network Upgrades included a 10% contingency.

2.3.6 Network Upgrade Alternatives Considered

All the ERIS network upgrades identified in Table 2.3.3-2 are direct upgrades of the ERIS thermal constraint facilities to ATC design standards and considered as least-cost solutions. Therefore, no other alternatives were examined for those solutions.

2.3.7 Potential Operating Restriction

The purpose of the study is to identify potential steady-state operating restrictions for study generators under prior outage conditions and raise awareness of these potential operating restrictions to customers. Real-time steady-state thermal constraints due to NERC Category P6 events (N-1-1) will be mitigated in the day-ahead and real-time market through the MISO binding constraint and other operating procedures. The study was performed on study models with final ERIS Network Upgrades and NRIS Network Upgrades included. Based on ATC generator operating restriction study methodology as described in Appendix A, the worst potential steady-state operating restrictions for study generators were identified and summarized in Table 2.3.7-1. If multiple P6 (N-1-1) events resulted in the same MW restriction, only the worst N-1-1 event (highest loading % on the constraint) was listed. The full NERC Category P6 steady-state study results were documented in Appendix F. Operating restrictions could occur if either of the contingent elements are out of service.

Table 2.3.7-1 – Worst Steady-State Operating Restrictions with ERIS and NRIS Network Upgrades Included

MISO Queue #	Model	Worst NERC Category P6 (N-1-1) Event		Max MW Allowed
		1 st Contingency	2 nd Contingency	
J986	23SH Batteries Discharging; 23SH Batteries Charging			0.00
J1000	23SH Batteries Discharging; 23SH Batteries Charging			0.00 ¹
J1002	23SUM Batteries Charging			50.04
J1003	-	-	-	-
J1042	-	-	-	-
J1101	23SH Batteries Charging			0.00 ²
J1153	-	-	-	-
J1154	23SH Batteries Discharging; 23SH Batteries Charging			0.00
J1171	-	-	-	-
J1183	-	-	-	-
J1188	-	-	-	-

¹ A higher rating on the constrained Ebenezer – Eden 138 kV line than what is currently in the study model is expected after Cardinal – Hickory Creek project which will likely reduce the restriction.

² J1101 would not be allowed to be in charging mode under this scenario.

3.0 STABILITY ANALYSIS

Stability analysis in Section 3 was performed using TSAT V19 to evaluate the stability performance of the generating facilities in the Apr 18 DPP WI Phase 3. Detailed study assumptions, criteria, and methodology are documented in Appendix A. For generators that require Electromagnetic transient (EMT) stability analysis using Power Systems Computer-Aided Design (PSCAD), results are documented in Section 6 of the report.

Stability plots consist of generator rotor angles, generator real power output, generator reactive power output, generator terminal voltages, and transmission bus voltages for each simulation. Simulations were performed with a 9-cycle flat start followed by the appropriate disturbance. Simulations were run for a 20-second duration.

3.1 Model Development

3.1.1 Study Case

A stability study case representing the 2023 shoulder load condition was developed by MISO and reviewed by ATC and Interconnection Customers. The model was developed and studied in TSAT v19. TSAT, like PSSE, is a root mean squared (RMS) based program that only models a network's positive sequence quantities. All units under study were dispatched to 100% of P_{MAX}. Additionally, the following local units were also dispatched to 100% of P_{MAX} to meet ATC Planning Criteria:

- J732
- J818
- J831
- J849
- J850
- J855
- J864
- J870/J871
- J878
- J886
- J928
- J947
- Columbia 1 and 2
- J390 CT1, CT2, and ST
- Riverside CT1, CT2, and ST
- Christiana 1, 2, and 3
- Glacier Hills Wind Farm
- Quiltblock Wind Farm
- Concord 1, 2, 3, and 4
- Elm Road 1 and 2
- Point Beach 1 and 2
- Weston 2, 3, 4, 31, and 32
- Oak Creek 5, 6, 7, and 8
- Paris 1, 2, 3, and 4
- South Fond du Lac 1, 2, 3, and 4

3.1.2 Benchmark Case

The Apr 18 DPP WI Phase 3 generating facilities were removed from the study case. The MISO Classic methodology, where generation was scaled in proportion to P_{max} minus P_{gen}, was used for power balance.

Faults were simulated using the 2023 shoulder study case. If a transient stability criteria constraint was identified, the same disturbance was repeated in the benchmark case.

3.2 J986 and J1002 Stability Study

3.2.1 J986 and J1002 Stability Results

Complete fault definitions and stability results for J986 and J1002 can be found in Appendix G.

Under all faults for J986 and J1002, the simulations show the system meeting all transient stability criteria.

3.2.2 Network Upgrades Identified in the Stability Study

Stability Network Upgrades are not required for the interconnection of the J986 and J1002 generating facilities.

3.2.3 Operating Restrictions Identified in the Stability Study

No stability operating restrictions were identified for the J986 and J1002 generating facilities.

3.2.4 PRC-024-2 Analyses

PRC-024 Analysis was completed for each generator using the methodology outlined in Appendix A section A.2.5 and the customer provided frequency and voltage trip settings. Results for J986 and J1002 are provided in tables below.

Table 3.2.4-1 – J986 PRC-024-2 Voltage Protection Relay Settings Results

Relay Type	Voltage Threshold (p.u.)	Relay Time (Sec)	Ride Through PRC-024-2 “No Trip Zone”?
Low Voltage	0.0	1	Yes
Low Voltage	0.45	5	Yes
Low Voltage	0.6	10	Yes
Low Voltage	0.7	20	Yes
High Voltage	2	0.16	Yes
High Voltage	1.25	1	Yes
High Voltage	1.2	2	Yes
High Voltage	1.16	3	Yes

Table 3.2.4-2 – J1002 PRC-024-2 Voltage Protection Relay Settings Results

Relay Type	Voltage Threshold (p.u.)	Relay Time (Sec)	Ride Through PRC-024-2 “No Trip Zone”?
Low Voltage	0.10	0.6	Yes
Low Voltage	0.50	1.9	Yes
High Voltage	1.15	3.0	Yes
High Voltage	1.40	0.0	Yes

Table 3.2.4-3 – J986 PRC-024-2 Frequency Protection Relay Settings Results

Relay Type	Frequency Threshold (Hz)	Relay Time (Sec)	Minimum Required Relay Time at This Frequency (Sec)	Ride Through PRC-024-2 “No Trip Zone”?
Low Frequency	55	0.16	0.0	Yes
Low Frequency	57	650	0.0	Yes
Low Frequency	57.5	5000	0.0	Yes
High Frequency	60.6	5000	429.5	Yes
High Frequency	62	350	0.0	Yes
High Frequency	63	350	0.0	Yes
High Frequency	65	0.16	0.0	Yes

Table 3.2.4-4 – J1002 PRC-024-2 Frequency Protection Relay Settings Results

Relay Type	Frequency Threshold (Hz)	Relay Time (Sec)	Minimum Required Relay Time at This Frequency (Sec)	Ride Through PRC-024-2 “No Trip Zone”?
Low Frequency	56.5	0.0	0.0	Yes
Low Frequency	57.5	10.0	0.0	Yes
Low Frequency	61.5	30.0	21.0	Yes
High Frequency	62.5	0.0	0.0	Yes

3.3 J1000 Stability Study

3.3.1 J1000 Stability Results

Complete fault definitions and stability results for J1000 can be found in Appendix G.

Under all faults for J1000, the simulations show the system meeting all transient stability criteria.

3.3.2 Network Upgrades Identified in the Stability Study

Stability Network Upgrades are not required for the interconnection of the J1000 generating facility.

3.3.3 Operating Restrictions Identified in the Stability Study

No stability operating restrictions were identified for the J1000 generating facility.

3.3.4 PRC-024-2 Analyses

PRC-024 Analysis was completed for each generator using the methodology outlined in Appendix A section A.2.5 and the customer provided frequency and voltage trip settings. Results for J1000 are provided in tables below.

Table 3.3.4-1 – J1000 PRC-024-2 Voltage Protection Relay Settings Results

Relay Type	Voltage Threshold (p.u.)	Relay Time (Sec)	Ride Through PRC-024-2 “No Trip Zone”?
Low Voltage	0.10	0.6	Yes
Low Voltage	0.50	1.9	Yes
High Voltage	1.15	3.0	Yes
High Voltage	1.40	0.0	Yes

Table 3.3.4-2 – J1000 PRC-024-2 Frequency Protection Relay Settings Results

Relay Type	Frequency Threshold (Hz)	Relay Time (Sec)	Minimum Required Relay Time at This Frequency (Sec)	Ride Through PRC-024-2 “No Trip Zone”?
Low Frequency	56.5	0.0	0.0	Yes
Low Frequency	57.5	10.0	0.0	Yes
Low Frequency	61.5	30.0	21.0	Yes
High Frequency	62.5	0.0	0.0	Yes

3.4 J1003 Stability Study

3.4.1 J1003 Stability Results

Complete fault definitions and stability results for J1003 can be found in Appendix G.

Under all faults for J1003, the simulations show the system meeting all stability criteria.

3.4.2 Network Upgrades Identified in the Stability Study

Stability Network Upgrades are not required for the interconnection of the J1003 generating facility.

3.4.3 Operating Restrictions Identified in the Stability Study

No stability operating restrictions were identified for the J1003 generating facilities.

3.4.4 PRC-024-2 Analyses

PRC-024 Analysis was completed for each generator using the methodology outlined in Appendix A section A.2.5 and the customer provided frequency and voltage trip settings. Results for J1003 are provided in tables below.

Table 3.4.4-1 – J1003 PRC-024-2 Voltage Protection Relay Settings Results

Relay Type	Voltage Threshold (p.u.)	Relay Time (Sec)	Ride Through PRC-024-2 “No Trip Zone”?
Low Voltage	0.10	0.6	Yes
Low Voltage	0.50	1.9	Yes
High Voltage	1.15	3.0	Yes
High Voltage	1.40	0.0	Yes

Table 3.4.4-2 – J1003 PRC-024-2 Frequency Protection Relay Settings Results

Relay Type	Frequency Threshold (Hz)	Relay Time (Sec)	Minimum Required Relay Time at This Frequency (Sec)	Ride Through PRC-024-2 “No Trip Zone”?
Low Frequency	56.5	0.0	0.0	Yes
Low Frequency	57.5	10.0	0.0	Yes
Low Frequency	61.5	30.0	21.0	Yes
High Frequency	62.5	0.0	0.0	Yes

3.5 J1042 Stability Study

3.5.1 J1042 Stability Results

Complete fault definitions and stability results for J1042 can be found in Appendix G

Under all faults for J1042, the simulations show the system meeting all transient stability criteria.

3.5.2 Network Upgrades Identified in the Stability Study

Stability Network Upgrades are not required for the interconnection of J1042 generating facility.

3.5.3 Operating Restrictions Identified in the Stability Study

No stability operating restrictions were identified for the J1042 generating facilities.

3.5.4 PRC-024-2 Analyses

PRC-024 Analysis was completed for each generator using the methodology outlined in Appendix A section A.2.5 and the customer provided frequency and voltage trip settings. Results for J1042 are provided in tables below.

Table 3.5.4-1 – J1042 PRC-024-2 Voltage Protection Relay Settings Results

Relay Type	Voltage Threshold (p.u.)	Relay Time (Sec)	Ride Through PRC-024-2 “No Trip Zone”?
Low Voltage	0.001	0.2	Yes
Low Voltage	0.43	0.4	Yes
Low Voltage	0.63	2.1	Yes
Low Voltage	0.73	3.1	Yes
Low Voltage	0.89	10	Yes
High Voltage	1.11	10	Yes
High Voltage	1.18	1.1	Yes
High Voltage	1.20	0.6	Yes
High Voltage	1.24	0.25	Yes
Low Voltage	0.001	0.2	Yes

Table 3.5.4-2 – J1042 PRC-024-2 Frequency Protection Relay Settings Results

Relay Type	Frequency Threshold (Hz)	Relay Time (Sec)	Minimum Required Relay Time at This Frequency (Sec)	Ride Through PRC-024-2 “No Trip Zone”?
Low Frequency	57.0	0.075	0.0	Yes
High Frequency	72.0	0.075	0.0	Yes

3.6 J1101 Stability Study

3.6.1 J1101 Stability Results

Complete fault definitions and stability results for J1101 can be found in Appendix G.

Under all faults for J1101, the simulations showed the system meeting all transient stability criteria with the generic model.

3.6.2 Network Upgrades Identified in the Stability Study

Stability Network Upgrades are not required for the interconnection of the J1101 generating facility with the generic model.

3.6.3 Operating Restrictions Identified in the Stability Study

No stability operating restrictions were identified for the J1101 generating facilities.

3.6.4 PRC-024-2 Analyses

PRC-024 Analysis was completed for each generator using the methodology outlined in Appendix A section A.2.5 and the customer provided frequency and voltage trip settings. Results for J1101 are provided in tables below. The tests were completed with the battery discharging and charging.

Table 3.6.4-1 – J1101 PRC-024-2 Voltage Protection Relay Settings Results

Relay Type	Voltage Threshold (p.u.)	Relay Time (Sec)	Ride Through PRC-024-2 “No Trip Zone”? (discharge/charge)
Low Voltage	0.45	1.0	Yes/Yes
Low Voltage	0.60	5.0	Yes/Yes
Low Voltage	0.70	2.00	Yes/Yes
Low Voltage	0.88	3.00	Yes/Yes
High Voltage	1.12	5.0	Yes/Yes
High Voltage	1.16	3.0	Yes/Yes
High Voltage	1.18	2.0	Yes/Yes
High Voltage	1.20	1.0	Yes/Yes
High Voltage	1.4	0.16	Yes/Yes

Table 3.6.4-2 – J1101 PRC-024-2 Frequency Protection Relay Settings Results

Relay Type	Frequency Threshold (Hz)	Relay Time (Sec)	Minimum Required Relay Time at This Frequency (Sec)	Ride Through PRC-024-2 “No Trip Zone”?
Low Frequency	57.0	0.01	0.0	Yes
Low Frequency	63.0	0.01	0.0	Yes

3.7 J1153 Stability Study

3.7.1 J1153 Stability Results

Complete fault definitions and stability results for J1153 can be found in Appendix G. The model did not include the required cap bank that was found in the FERC 827 analysis. This was forcing the generator to its maximum var output and when a fault [REDACTED] was applied, the voltage at the terminals would go high resulting in tripping. After the cap bank was added, the var output at the generator returned to near zero and the terminal voltages would be within a normal range. This resulted in no tripping of the generator.

Under all faults for J1153, with the required cap bank, the simulations showed the system meeting all transient stability criteria with the generic model.

3.7.2 Network Upgrades Identified in the Stability Study

Stability Network Upgrades are not required for the interconnection of the J1153 generating facility with the generic model.

3.7.3 Operating Restrictions Identified in the Stability Study

No stability operating restrictions were identified for the J1153 generating facilities.

3.7.4 PRC-024-2 Analyses

PRC-024 Analysis was completed for each generator using the methodology outlined in Appendix A section A.2.5 and the customer provided frequency and voltage trip settings. Results for J1153 are provided in Tables below.

Table 3.7.4-1 – J1153 PRC-024-2 Voltage Protection Relay Settings Results

Relay Type	Voltage Threshold (p.u.)	Relay Time (Sec)	Ride Through PRC-024-2 “No Trip Zone”?
Low Voltage	0.50	2.0	Yes
Low Voltage	0.70	10.0	Yes
Low Voltage	0.880	20.0	No
High Voltage	1.100	10.000	Yes
High Voltage	1.200	1.0	Yes
High Voltage	1.300	0.050	Yes

Table 3.7.4-2 – J1153 PRC-024-2 Frequency Protection Relay Settings Results

Relay Type	Frequency Threshold (Hz)	Relay Time (Sec)	Minimum Required Relay Time at This Frequency (Sec)	Ride Through PRC-024-2 “No Trip Zone”?
Low Frequency	57.0	0.5	0.0	Yes
Low Frequency	58.3	600	14.9	Yes
High Frequency	60.7	600	307.0	Yes
High Frequency	62.0	0.5	0.0	Yes

3.8 J1154 Stability Study

3.8.1 J1154 Stability Results

Complete fault definitions and stability results for J1154 can be found in Appendix G.

Under all other faults for J1154, the simulations show the system meeting all stability criteria.

3.8.2 Network Upgrades Identified in the Stability Study

Stability Network Upgrades are not required for the interconnection of the J1154 generating facility.

3.8.3 Operating Restrictions Identified in the Stability Study

No stability operating restrictions were identified for the J1154 generating facilities.

3.8.4 PRC-024-2 Analysis

PRC-024 Analysis was completed for each generator using the methodology outlined in Appendix A section A.2.5 and the customer provided frequency and voltage trip settings. Results for J1154 are provided in tables below.

Table 3.8.4-1 – J1154 PRC-024-2 Voltage Protection Relay Settings Results

Relay Type	Voltage Threshold (p.u.)	Relay Time (Sec)	Ride Through PRC-024-2 “No Trip Zone”?
Low Voltage	0.50	2.0	Yes
Low Voltage	0.70	10.0	Yes
Low Voltage	0.880	20.0	No
High Voltage	1.100	10.000	Yes
High Voltage	1.200	1.0	Yes
High Voltage	1.300	0.050	Yes

Table 3.8.4-2 – J1154 PRC-024-2 Frequency Protection Relay Settings Results

Relay Type	Frequency Threshold (Hz)	Relay Time (Sec)	Minimum Required Relay Time at This Frequency (Sec)	Ride Through PRC-024-2 “No Trip Zone”?
Low Frequency	57.0	0.5	0.0	Yes
Low Frequency	58.3	600	14.9	Yes
High Frequency	60.7	600	307.0	Yes
High Frequency	62.0	0.5	0.0	Yes

3.9 J1171 Stability Study

3.9.1 J1171 Stability Results

Complete fault definitions and stability results for J1171 can be found in Appendix G. Under all faults for J1171, the simulations show the system meeting all stability criteria.

3.9.2 Network Upgrades Identified in the Stability Study

Stability Network Upgrades are not required for the interconnection of the J1171 generating facility.

3.9.3 Operating Restrictions Identified in the Stability Study

No stability operating restrictions were identified for the J1171 generating facilities.

3.9.4 PRC-024-2 Analyses

PRC-024 Analysis was completed for each generator using the methodology outlined in Appendix A section A.2.5 and the customer provided frequency and voltage trip settings. Results for J1171 are provided in tables below.

Table 3.9.4-1 – J1171 PRC-024-2 Voltage Protection Relay Settings Results

Relay Type	Voltage Threshold (p.u.)	Relay Time (Sec)	Ride Through PRC-024-2 “No Trip Zone”?
Low Voltage	0.500	10.000	Yes
Low Voltage	0.700	21.000	Yes
Low Voltage	0.900	30.000	Yes
High Voltage	1.110	30.000	Yes
High Voltage	1.200	2.000	Yes
High Voltage	1.25	0.200	Yes

Table 3.9.4-2 – J1171 PRC-024-2 Frequency Protection Relay Settings Results

Relay Type	Frequency Threshold (Hz)	Relay Time (Sec)	Minimum Required Relay Time at This Frequency (Sec)	Ride Through PRC-024-2 “No Trip Zone”?
Low Frequency	56.5	6.0	0.0	Yes
High Frequency	63.5	6.0	0.0	Yes

3.10 J1183 Stability Study

3.10.1 J1183 Stability Results

MISO ran the stability assessment in TSAT for this cycle. ATC does not have a TSAT model for the HVDC device at Mackinac. As a result, any TSAT study in the U.P. would be deficient. J1183 was studied with PSCAD and the stability results from that model was used for this cycle. Under all faults for J1183, the simulations show the system meeting all stability criteria.

3.10.2 Network Upgrades Identified in the Stability Study

Stability Network Upgrades are not required for the interconnection of the J1183 generating facility.

3.10.3 Operating Restrictions Identified in the Stability Study

No stability operating restrictions were identified for the J1183 generating facilities.

3.10.4 PRC-024-2 Analyses

PRC-024 Analysis and Momentary Cessation analysis was completed in the PSCAD study and no violations were found.

3.11 J1188 Stability Study

3.11.1 J1188 Stability Results

Complete fault definitions and stability results for J1188 can be found in Appendix G.

Under all faults for J1188, the simulations show the system meeting all stability criteria.

3.11.2 Network Upgrades Identified in the Stability Study

Stability Network Upgrades are not required for the interconnection of the J1188 generating facility.

3.11.3 Operating Restrictions Identified in the Stability Study

No stability operating restrictions were identified for the J1188 generating facilities.

3.11.4 PRC-024-2 Analysis

PRC-024 Analysis was completed for each generator using the methodology outlined in Appendix A section A.2.5 and the customer provided frequency and voltage trip settings. Results for J1188 are provided in tables below.

Table 3.11.4-1 – J1188 PRC-024-2 Voltage Protection Relay Settings Results

Relay Type	Voltage Threshold (p.u.)	Relay Time (Sec)	Ride Through PRC-024-2 “No Trip Zone”?
Low Voltage	0.500	10.000	Yes
Low Voltage	0.700	21.000	Yes
Low Voltage	0.900	30.000	Yes
High Voltage	1.110	30.000	Yes
High Voltage	1.200	2.000	Yes
High Voltage	1.25	0.200	Yes

Table 3.11.4-2 – J1188 PRC-024-2 Frequency Protection Relay Settings Results

Relay Type	Frequency Threshold (Hz)	Relay Time (Sec)	Minimum Required Relay Time at This Frequency (Sec)	Ride Through PRC-024-2 “No Trip Zone”?
Low Frequency	56.5	6.0	0.0	Yes
High Frequency	63.5	6.0	0.0	Yes

3.12 Summary of Operating Restriction Identified in the Stability Analysis

For NERC Category P6 stability constraints, operating restrictions may be required during prior outages to avoid instability caused by the next event. There were no P6 operating restrictions for study generators found in the Apr 18 DPP WI Phase 3 studies.

3.13 Network Upgrades Identified in the Stability Analysis

There are no network upgrades identified in the TSAT stability Analysis for Phase 3.

4.0 SHORT CIRCUIT ANALYSIS

Short Circuit analysis was performed in Phase 3 with the final ERIS and NRIS network upgrade projects included in ATC Protection CAPE (Computer-Aided

Protection Engineering) models, which include all ATC-owned substations, all ATC local distribution customer owned substations, as well as limited buses into ATC neighboring systems (MP, ComEd, XCEL, DPC, ITC). The analysis was performed according to ATC short circuit analysis methodology as described in Appendix A. Short circuit analysis results for facilities under MISO Functional Control are described in this section. For each substation under MISO functional control that shows a bus having fault current (SLG) increased by 10% or more, ATC will perform detailed grounding grid analysis if in-house grounding grid modeling is available at the time of the study. Only the short circuit results for the substations that do not pass the detailed grounding analysis (i.e. not meeting the safety criteria) are provided.

Facilities that are not under MISO Functional Control are considered as Affected System. Section 7.1 provides the substation ground grid analysis results on Affected System modelled in ATC Protection CAPE models.

4.1 Short Circuit Analysis Results

Maximum and minimum fault duty was calculated at the POI for each generator request and results are summarized in Table 4.1-1, Table 4.1-2, Table 4.1-3 and Table 4.1-4.

Table 4.1-1 – Outage Assumptions for Minimum Fault Duty Calculations

MISO Queue #	Contingency
J986	[REDACTED]
J1000	[REDACTED]
J1002	[REDACTED]
J1003	[REDACTED]
J1042	[REDACTED]
J1101	[REDACTED]
J1153	[REDACTED]
J1154	[REDACTED]
J1171	[REDACTED]
J1183	[REDACTED]
J1188	[REDACTED]

Table 4.1-2 – Maximum and Minimum Fault Duty at POIs

Location	Maximum Fault Duty (Amps)		Minimum Fault Duty (Amps)	
	Single-Phase	Three-Phase	Single-Phase	Three-Phase
J986 POI without J986	6,386	7,652	1,577	2,171
J986 POI with J986	9,867	8,502	4,192	3,017
J1000 POI without J1000	6,463	8,573	4,626	5,694
J1000 POI with J1000	7,514	9,053	5,613	6,174
J1002 POI without J1002	4,712	6,012	2,939	3,750
J1002 POI with J1002	6,499	7,616	4,709	5,361
J1003 POI without J1003	7,743	7,448	2,765	4,283
J1003 POI with J1003	9,156	8,402	4,767	5,224
J1042 POI without J1042	10,173	12,877	6,329	7,364
J1042 POI with J1042	13,085	13,886	8,820	8,375
J1101 POI without J1101	19,579	18,715	11,212	11,054
J1101 POI with J1101	20,853	19,143	12,609	11,498
J1153 POI without J1153	4,462	6,510	1,087	1,490
J1153 POI with J1153	7,352	7,363	3,008	2,344
J1154 POI without J1154	10,045	13,688	5,520	7,708
J1154 POI with J1154	13,833	14,547	8,902	8,567
J1171 POI without J1171	7,064	9,405	1,965	3,025
J1171 POI with J1171	8,252	9,939	3,260	3,557
J1183 POI without J1183	5,001	4,572	3,433	2,840
J1183 POI with J1183	5,001	4,580	3,433	2,848
J1188 POI without J1188	4,804	7,443	2,852	4,279
J1188 POI with J1188	6,725	7,996	4,580	4,830

Table 4.1-3 – Maximum Fault Thevenin Equivalent Data

Location	Positive Sequence (Ohm)	Negative Sequence (Ohm)	Zero Sequence (Ohm)	Three- Phase Thevenin X/R Ratio	Single- Phase Thevenin X/R Ratio	Three- Phase ANSI 37.010 X/R Ratio	Single- Phase ANSI 37.010 X/R Ratio
J986 POI without J986	2.03993 + j10.2093	2.04420 + j10.2165	3.79975 + j16.1549	5.0	4.6	5.6	6.0
J986 POI with J986	2.03993 + j10.2093	2.04420 + j10.2165	0.76708 + j6.46468	5.0	5.5	5.6	6.7
J1000 POI without J1000	1.52716 + j9.55942	1.52934 + j9.55927	3.60427 + j17.7099	6.3	5.5	6.8	6.5
J1000 POI with J1000	1.52716 + j9.55942	1.52934 + j9.55927	2.38426 + j14.5033	6.3	6.2	6.8	8.3
J1002 POI without J1002	3.15658 + j12.8718	3.16112 + j12.8731	5.27576 + j23.5409	4.1	4.3	4.5	5.0
J1002 POI with J1002	2.02452 + j10.2616	2.02726 + j10.2627	2.65247 + j15.5804	5.1	5.4	10.2	9.5
J1003 POI without J1003	1.28603 + j5.19245	1.28806 + j5.19367	0.70746 + j4.64540	4.0	4.6	4.7	5.6
J1003 POI with J1003	1.28627 + j5.19290	1.28830 + j5.19412	0.54088 + j3.92997	4.0	4.6	4.7	5.5
J1042 POI without J1042	0.96670 + j6.36302	0.98223 + j6.42358	2.11939 + j10.8815	6.6	5.8	6.9	6.9
J1042 POI with J1042	0.96674 + j6.36314	0.98227 + j6.42369	0.99106 + j7.22275	6.6	6.8	6.9	8.6
J1101 POI without J1101	0.24127 + j4.25309	0.28201 + j4.24721	0.32485 + j3.65196	17.6	14.3	32.4	24.2
J1101 POI with J1101	0.24132 + j4.25359	0.28208 + j4.24771	0.33404 + j3.29990	17.6	13.8	32.4	23.5
J1153 POI without J1153	1.34215 + j12.1732	1.36607 + j12.1736	5.44470 + j28.5757	9.1	6.5	9.3	7.2
J1153 POI with J1153	1.34216 + j12.1733	1.36608 + j12.1737	0.96838 + j12.2257	9.1	9.9	9.3	12.0
J1154 POI without J1154	0.74883 + j5.77445	0.76394 + j5.80801	3.40497 + j11.6368	7.7	4.7	8.2	4.9
J1154 POI with J1154	0.70516 + j5.59964	0.71912 + j5.63127	1.06001 + j6.36284	7.9	7.1	9.4	10.8
J1171 POI without J1171	1.07550 + j8.40404	1.07723 + j8.41130	7.81918 + j15.4539	7.8	3.2	8.9	4.6
J1171 POI with J1171	1.07663 + j8.40836	1.07836 + j8.41563	5.12890 + j12.8682	7.8	4.1	8.9	7.2
J1183 POI without J1183	5.03672 + j24.7140	5.05217 + j24.6814	3.65001 + j20.0424	4.9	5.1	6.0	6.1
J1183 POI with J1183	5.03672 + j24.7140	5.05217 + j24.6814	3.65001 + j20.0424	4.9	5.1	6.0	6.1
J1188 POI without J1188	1.25846 + j5.20080	1.26488 + j5.19998	3.78212 + j13.5790	4.1	3.8	4.5	4.0
J1188 POI with J1188	1.31147 + j5.30254	1.31814 + j5.30162	1.65787 + j8.34701	4.0	4.4	4.5	5.1

Table 4.1-4 – Minimum Fault Thevenin Equivalent Data

Location	Positive Sequence (Ohm)	Negative Sequence (Ohm)	Zero Sequence (Ohm)	Three-Phase Thevenin X/R Ratio	Single-Phase Thevenin X/R Ratio	Three-Phase ANSI 37.010 X/R Ratio	Single-Phase ANSI 37.010 X/R Ratio
J986 POI without J986	9.18269 + j35.5377	9.18755 + j35.5385	18.8178 + j75.8478	3.9	4.0	4.5	4.6
J986 POI with J986	9.18269 + j35.5377	9.18755 + j35.5385	0.65039 + j9.36188	3.9	4.2	4.5	4.9
J1000 POI without J1000	2.38849 + j13.7924	2.38913 + j13.7925	4.54312 + j23.2035	5.8	5.4	6.3	6.5
J1000 POI with J1000	2.38849 + j13.7924	2.38913 + j13.7925	2.68472 + j17.9740	5.8	6.1	6.3	8.0
J1002 POI without J1002	4.72515 + j20.7156	4.73271 + j20.7157	7.94319 + j37.7306	4.4	4.5	4.8	5.4
J1002 POI with J1002	2.42828 + j14.6572	2.43194 + j14.6577	3.01782 + j20.6988	6.0	6.3	13.0	11.4
J1003 POI without J1003	2.93475 + j8.82719	2.93671 + j8.83014	7.67874 + j23.3889	3.0	3.0	3.2	3.1
J1003 POI with J1003	2.93568 + j8.82843	2.93764 + j8.83138	2.38159 + j12.3732	3.0	3.6	3.2	4.1
J1042 POI without J1042	1.65193 + j11.4592	1.68020 + j11.4667	2.66029 + j15.8635	6.9	6.5	7.2	7.5
J1042 POI with J1042	1.65204 + j11.4596	1.68031 + j11.4671	0.97813 + j9.09730	6.9	7.4	7.2	8.6
J1101 POI without J1101	0.53170 + j7.18383	0.56788 + j7.17720	0.55300 + j6.88108	13.5	12.9	31.9	26.0
J1101 POI with J1101	0.53192 + j7.18527	0.56811 + j7.17863	0.55205 + j5.66706	13.5	12.1	31.9	24.8
J1153 POI without J1153	6.23111 + j53.1777	6.25051 + j53.1692	18.1235 + j111454	8.5	7.1	8.8	8.5
J1153 POI with J1153	6.23132 + j53.1786	6.25073 + j53.1701	0.45019 + j17.7641	8.5	9.6	8.8	9.9
J1154 POI without J1154	1.42668 + j10.2365	1.44038 + j10.3560	6.36045 + j21.6505	7.2	4.6	7.6	4.8
J1154 POI with J1154	1.28380 + j9.70074	1.29458 + j9.80792	1.07747 + j8.46272	7.6	7.7	9.8	11.5
J1171 POI without J1171	4.24389 + j25.9978	4.26474 + j26.0138	15.3994 + j67.2060	6.1	5.0	6.2	5.1
J1171 POI with J1171	4.25764 + j26.0388	4.27857 + j26.0548	4.35282 + j33.1254	6.1	6.6	6.2	7.9
J1183 POI without J1183	8.11490 + j37.0291	8.13268 + j36.9905	5.02655 + j28.1734	4.6	4.8	4.8	4.5
J1183 POI with J1183	8.11490 + j37.0291	8.13268 + j36.9905	5.02655 + j28.1734	4.6	4.8	4.8	5.2
J1188 POI without J1188	2.45653 + j8.98030	2.46700 + j8.97926	6.81561 + j22.0926	3.7	3.4	3.9	3.6
J1188 POI with J1188	2.63918 + j9.28192	2.65040 + j9.28057	2.05971 + j10.9444	3.5	4.0	3.9	4.6

ATC breaker duty analysis results for over duty breakers are summarized in Table 4.1-5 and Table 4.1-6.

Table 4.1-5 – Over Duty Breakers Found in SLG Analysis

Breaker Name	Breaker Rating	Base Case SLG Margin	Study Case SLG Margin
None	N/A	N/A	N/A

Table 4.1-6 – Over Duty Breakers Found in 3PG Analysis

Breaker Name	Breaker Rating	Base Case 3PG Margin	Study Case 3PG Margin
None	N/A	N/A	N/A

Please note, no breaker duty analysis was performed for Affected System equipment. Appendix N provides short circuit data for all substation buses 69 kV and above included in ATC protection CAPE models that show a 1% increase in SLG or 3PG fault current. Any non-ATC equipment requiring a change would be considered Affected System.

Ground fault analysis results are summarized in Table 4.1-7.

Table 4.1-7 – Buses with 10% or More Fault Current Increase in SLG Analysis

SUBSTATION	Facility Owner	BUS	kV	BEFORE (AMPS)	AFTER (AMPS)	% Change
HANCOCK	Alliant	883300	69	3110.3	4230.2	36.01
HOLLYWOOD	Alliant	894800	138	9090.2	10510.6	15.63
ROEDER	Alliant	923400	138	4430.6	4903.4	10.67
CHAFFEE CREEK	ATC	831500	69	3892.2	4374.1	12.38
HOWARDS GROVE	ATC	922300	138	3036.3	3411.7	12.36
HOWARDS GROVE	ATC	922301	138	3036.3	3411.7	12.36
NORTH BEAVER DAM	ATC	822200	69	7740.5	9156	18.29
NORTH BEAVER DAM	ATC	917300	138	4148.1	4443.7	7.13
NORTH LAKE GENEVA	ATC	908500	138	10210.3	13085	28.15
NORTH LAKE GENEVA	ATC	908501	138	10210.3	13085	28.15
NORTH LAKE GENEVA	ATC	808300	69	8983.1	9429.8	4.97
PORT EDWARDS	ATC	835800	69	14630.3	15379.5	5.12
PORT EDWARDS	ATC	835900	69	14630.3	15379.5	5.12
PORT EDWARDS	ATC	924100	138	9764.3	11381.9	16.57
PORT EDWARDS	ATC	924101	138	9764.3	11381.9	16.57
PORT EDWARDS	ATC	924102	138	9764.3	11381.9	16.57
SAND LAKE	ATC	994000	69	3413.7	5816.5	70.39
SAND LAKE	ATC	924150	138	4248.7	5557.9	30.81
SAND LAKE	ATC	993900	138	4246.9	5554.6	30.79
SARATOGA	ATC	979700	115	10310.8	11196.4	8.59
SARATOGA	ATC	924000	138	9261.2	10709	15.63
SARATOGA	ATC	924001	138	9261.2	10709	15.63

SARATOGA	ATC	835200	69	4936	5060.6	2.52
SHEEPSKIN	ATC	910400	69	4804.1	6724.7	39.98
WAUTOMA	ATC	828600	69	6120.7	8352	36.45
WAUTOMA	ATC	923500	138	4106.2	6498.9	58.27
PLYMOUTH SUB 4	Plymouth Utilities	918800	138	3215.3	3784.6	17.71
BUTTERNUT	We Energies	927000	138	7063.5	8252.3	16.83
BUTTERNUT	We Energies	926900	138	7063.5	8252.3	16.83
HOLLAND	We Energies	928800	138	4461.4	7352.1	64.79
HOLLAND	We Energies	928801	138	4461.4	7352.1	64.79

4.2 Short Circuit Network Upgrade Identified

Short circuit Network Upgrades are summarized in Table 4.2-1. Details on cost estimating/allocation methodology and cost allocations are provided in Section 9.

Table 4.2-1 – Short Circuit Network Upgrade Identified

SUBSTATION	Facility Owner	Tenants for Joint Use Substations	Total Cost (\$)
HANCOCK	Alliant	ATC	220,000
HOLLYWOOD	Alliant	ATC	220,000
ROEDER	Alliant	ATC	220,000
SAND LAKE	ATC	-	220,000
WAUTOMA	ATC	Alliant	220,000
SHEEPSKIN	ATC	Alliant	220,000
NORTH LAKE GENEVA	ATC	Alliant	220,000
NORTH BEAVER DAM	ATC	Alliant	220,000
PORT EDWARDS	ATC	Alliant	220,000
SARATOGA	ATC	Alliant	220,000
CHAFFEE CREEK	ATC	-	220,000
HOWARDS GROVE	ATC	Alliant	220,000
PLYMOUTH SUB 4	Plymouth Utilities	ATC	220,000
HOLLAND	We Energies	ATC	220,000
BUTTERNUT	We Energies	ATC	220,000

5.0 TRANSFORMER ENERGIZATION ANALYSIS

5.1 Transformer Energization Study Results

Transformer Energization Analysis were performed based on ATC transformer initial energization criteria as described in Appendix A. The 2023 shoulder model (batteries charging) was used with all identified Phase 3 ERIS and NRIS Network Upgrades included. The results are summarized in Table 5.1-1. No constraints (Vmin inrush below 70%) were found.

Table 5.1-1 – Inrush Calculations Using 2023 Shoulder Battery Charging Model

Generation Project	Transformer(s)		PSSE POI Bus #	Fault Current (A)	Vmin Inrush (pu)	
	Number	Windings Modelled in PSSE			Raw	With Multiplier
J986	1	2	89864	6373	0.738	0.849
J1000	1	2	40004	7612	0.929	1.068
J1002	1	2	699235	5407	0.828	0.952
J1003	1	2	698222	7124	0.847	0.974
J1042	1	3	699085	11162	0.821	0.945
J1101	1	2	699620	19755	0.985	1.132
J1153	1	2	699288	6355	0.798	0.918
J1154	1	2	699340	11878	0.915	1.052
J1171	1	2	699270	8628	0.757	0.871
J1183	1	2	693701	3206	0.818	0.940
J1188	1	2	699104	7216	0.859	0.987

6.0 PSCAD ANALYSIS

ATC performed a low short circuit strength screening analysis for all inverter-based resources in the DPP-2018-April East (ATC) Area Phase 1 which determined that the following requests that have proceeded to Phase 3 were connecting to relatively weaker portions of the ATC system that would require further electromagnetic transient (EMT) analysis in PSCAD:

- J986
- J1000
- J1002
- J1042
- J1153
- J1183

J1042 was considered to be connecting to a weak part of the system in Phase 1 when also considering requests J1044, J1047, J1010, and J1011. Since all four of those requests had withdrawn before Phase 3 it was determined that the local system would be sufficiently strong for J1042 on it's own and would not need to be studied in PSCAD in Phase 3.

PSCAD requires much more data and processing time than traditional steady state and dynamics software packages used in transmission planning such that system wide area models do not exist. To study the remaining five generators connecting to weaker portions of the ATC system, four different local models were produced including:

- Southwest Wisconsin (SW WI) Model
- J1153 Model
- J986/J1002 Model
- J1183 Model

6.1 SW WI PSCAD Study

Full details of the SW WI PSCAD study can be found in Appendix L.

6.1.1 SW WI PSCAD Study Results

All events studied were found to be stable for all elements monitored. Full results of the SW WI PSCAD study are summarized in Table 6.1.1-1 below. Full detailed results for all PSCAD results are included in Appendix L- SW WI PSCAD Report.

[illegible]

6.1.2 Network Upgrades Identified in the SW WI PSCAD Study

No network upgrades were identified to mitigate constraints from the Phase 3 SW WI PSCAD study.

6.1.3 Operating Restrictions Identified in the SW WI PSCAD Study

No operating restrictions were identified to mitigate constraints from the Phase 3 SW WI PSCAD study.

6.2 J1153 PSCAD Study

Full details of the J1153 PSCAD study can be found in Appendix L.

6.2.1 J1153 PSCAD Study Results

All events studied were found to be stable for all elements monitored. Full results of the J1153 PSCAD study are summarized in Table 6.2.1-1 below. Full detailed results for all PSCAD results are included in Appendix L – J1153 PSCAD Report.

Table 6.2.1-1 – J1153 PSCAD Results

TPL	Fault Description	Fault Type	Stability Performance
			J1153
P1.2	[REDACTED]	3PG	OK
		1PG	OK
P4.5	[REDACTED]	3PG	OK
P6	[REDACTED]	3PG	OK
		1PG	OK
P6	[REDACTED]	3PG	OK
		1PG	OK

6.2.2 Network Upgrades Identified in the J1153 PSCAD Study

No network upgrades were identified to mitigate constraints from the Phase 3 J1153 PSCAD study.

6.2.3 Operating Restrictions Identified in the J1153 PSCAD Study

No operating restrictions were identified to mitigate constraints from the Phase 3 J1153 PSCAD study.

6.3 J986/J1002 PSCAD Study

Full details of the J986/J1002 PSCAD study can be found in Appendix L.



6.3.1 J986/J1002 PSCAD Study Results

The J986/J1002 analysis found that J986 trips on the contingency of [REDACTED]. This fault was rerun with all of the protection settings disabled and the response was also found to be undamped so operating restrictions were determined to mitigate this event.

The J986/J1002 analysis found that J986 and J1002 were undamped over the last two seconds of the simulation for the contingency of [REDACTED]. Operating restrictions were derived to mitigate this constraint and are detailed in Section 6.3.3.

All other events studied were found to be stable for all elements monitored. Full results of the J986/J1002 PSCAD study are summarized in Table 6.3.1-1 below. Full detailed results for all PSCAD results are included in Appendix L- J986_J1002_PSCAD Report.

Table 6.3.1-1 – J986/J1002 PSCAD Results

TPL	Fault Description	Fault Type	Stability Performance	
			J986	J1002
P1.2	[REDACTED]	3PG	OK	OK
		1PG	OK	OK
P4.5	[REDACTED]	1PG	OK	OK
P6	[REDACTED]	3PG	J986 trips	J986 trips
		1PG	J986 trips	J986 trips
P6	[REDACTED]	3PG	undamped	undamped
		1PG	undamped	undamped
P6	[REDACTED]	3PG	OK	OK
		1PG	OK	OK

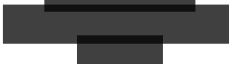
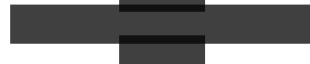
6.3.2 Network Upgrades Identified in the J986/J1002 PSCAD Study

No network upgrades were identified to mitigate constraints from the Phase 3 J986/J1002 PSCAD study.

6.3.3 Operating Restrictions Identified in the J986/J1002 PSCAD Study

For P6 stability constraints operating restrictions may be required under prior outages conditions to avoid instability caused by the next event. The maximum allowed real power outputs after prior outages were identified for the J986 and J1002 generating facilities to mitigate P6 stability constraints. See Table 6.3.3-1 below.

Table 6.3.3-1 – J986/J1002 PSCAD Identified Operating Restrictions

Event	Prior Outage	Faulted Element	Maximum Real Power Output of J986/J1002
			To Maintain IBR Stability and Other Voltage and Oscillation Criteria (MW)
P6			100 (J986 restriction only)
P6			200 (J986/J1002 combined restriction)

6.4 J1183 PSCAD Study

Full details of the J1183 PSCAD study can be found in Appendix L.

6.4.1 J1183 PSCAD Study Results

All events studied were found to be stable for all elements monitored.

6.4.2 Network Upgrades Identified in the J1183 PSCAD Study

No network upgrades were identified to mitigate constraints from the Phase 3 J1183 PSCAD study.

6.4.3 Operating Restrictions Identified in the J1183 PSCAD Study

No operating restrictions were identified to mitigate constraints from the Phase 3 J1183 PSCAD study.

6.5 Summary of Operating Restriction Identified in the PSCAD Analysis

For NERC Category P6 stability constraints, operating restrictions may be required under prior outages conditions to avoid instability caused by the next event. A summary of all P6 operating restrictions found in the PSCAD analysis for study generators in Apr 18 DPP WI Phase 3 is provided in Table 6.5-1 below. Maximum allowed real power outputs after prior outages were identified for generating facility to mitigate P6 stability constraints.

Table 6.5.1-1 – Summary of All PSCAD Identified Operating Restrictions

MISO Queue #	Max MW	Worst NERC Category P6 (N-1-1) Event		PSCAD Analysis Max MW Allowed
		1st Contingency	2nd Contingency	
J986	150			100
J986/J1002	249			200

7.0 AFFECTED SYSTEM ANALYSIS

This section contains two study efforts on Affected System. Section 7.1 provides the substation ground grid analysis results on Affected System modelled in ATC Protection CAPE models. The analysis was performed in Phase 3 with the final Phase 3 ERIS and NRIS Network Upgrade projects included. Section 7.2 provides PJM Affected System steady state analysis results.

7.1 Affected System Substation Ground Grid Analysis

7.1.1 Affected System Substation Ground Grid Analysis Results

Some Affected System facilities that are not under MISO functional control are modeled in the ATC Protection CAPE models and were evaluated in the short circuit analysis. Affected System substation single-line-ground (SLG) fault analysis results are summarized in Table 7.1.1-1.

Table 7.1.1-1 – Affected System Buses with 10% or More Fault Current Increase in SLG Analysis

SUBSTATION	Facility Owner	BUS	kV	BEFORE (AMPS)	AFTER (AMPS)	% Change
ACEC HANCOCK	Adams-Columbia Coop	831100	69	3099.4	4210	35.83
ACEC RICHFORD	Adams-Columbia Coop	831000	69	3650.2	4076.3	11.67
ACEC WAUTOMA	Adams-Columbia Coop	829500	69	3863.5	4499.6	16.46
ACEC WILD ROSE PUMP	Adams-Columbia Coop	993800	69	3316	3668.9	10.64
BEAVER DAM 3RD ST	Alliant	822300	69	6171.2	7000.8	13.44
CENTER ST	Alliant	822400	69	5990.6	6694.2	11.75
COLOMA	Alliant	883200	69	3217	3795.3	17.98
DANA CORP	Alliant	810800	69	4154.1	5367	29.20
PLAINFIELD (ALTE)	Alliant	831400	69	2266.9	3117.9	37.54
SAUNDERS CREEK	Alliant	810600	69	4510.6	5661.8	25.52
SILVER LAKE (ALTE)	Alliant	828400	69	4174.9	4912.9	17.68
VULCAN CHEM	Alliant	836100	138	9504.2	11056	16.33
VULCAN CHEM	Alliant	836101	138	9504.2	11056	16.33
REC FULTON	Rock Energy Coop	355100	69	3895.9	4312.7	10.70

Please note, no breaker duty analysis was performed for Affected System equipment. Appendix N provides short circuit data for all substation buses 69 kV and above included in ATC protection CAPE models that show a 1% increase in SLG or 3PG fault current. Any non-ATC equipment requiring a change would be considered Affected System.

7.1.2 Potential Affected System Substation Ground Grid Upgrade Identified

Potential Affected System substation ground grid upgrades are summarized in Table 7.1.2-1. Details on cost allocation methodology and cost allocation percentages are provided in Section 9.

Table 7.1.2-1 – Potential Affected System Substation Ground Grid Upgrade Identified

SUBSTATION	Facility Owner	Total Cost (\$)
ACEC HANCOCK	Adams-Columbia Coop	To be determined by Affected System Facility Owner
ACEC WAUTOMA	Adams-Columbia Coop	
ACEC RICHFORD	Adams-Columbia Coop	
ACEC WILD ROSE PUMP	Adams-Columbia Coop	
PLAINFIELD (ALTE)	Alliant	
DANA CORP	Alliant	
SAUNDERS CREEK	Alliant	
SILVER LAKE (ALTE)	Alliant	
VULCAN CHEM	Alliant	
BEAVER DAM 3RD ST	Alliant	
COLOMA	Alliant	
CENTER ST	Alliant	
REC FULTON	Rock Energy Coop	

7.2 PJM Affected System Steady State Contingency Analysis

PJM performed the PJM affected system analysis and the report can be found in Appendix H. There are no PJM Affected System Network Upgrades that have cost allocations to the study generators in Apr 18 DPP Phase 2 group.

8.0 DELIVERABILITY STUDY

Generator interconnection requests have to pass Generator Deliverability Study to be granted Network Resource Interconnection Services (NRIS). If the generator is determined as not fully deliverable, the customer can either choose to elect the amount of NRIS available without upgrades or build system upgrades that will make the generator fully deliverable. Generator Deliverability Study ensures that the Network Resources, on an aggregate basis, can meet the MISO aggregate load requirements during system peak condition without getting bottlenecked up.

MISO Generator Deliverability Study methodology is described in MISO BPM-15.

8.1 Study Summary

The summary of MISO deliverability results based on the 2023 summer peak study model is shown in the following tables.

Table 8.1-1 below lists the deliverability results with ERIIS Network Upgrades included in the NRIS analysis. Minimum NR Deliverable is the amount of service to be granted if none of the projects contributing to this constraint, reduced service levels or withdrew. Maximum NR Deliverable is the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared deduction.

Table 8.1-1 – NRIS Analysis Summary

MISO Queue #	Area	NR Tested	Minimum NR Deliverable (MW)	Maximum NR Deliverable (MW)
J986	ATC	149.76	149.76	149.76
J1000	ATC	50	50	50
J1002	ATC	99	99	99
J1003	ATC	50	0	17.1
J1042	ATC	180	142.97	147
J1101	ATC	20	20	20
J1153	ATC	150	150	150
J1154	ATC	75	0	11.84
J1171	ATC	100	34.16	47.38
J1183	ATC	0	0	0
J1188	ATC	50	0	35.28

Table 8.1-2 below lists all of the NRIS constraints from the deliverability study and the identified NRIS Network Upgrades. Both ERIIS Network Upgrades and NRIS Network Upgrades must be made for 100% NRIS, i.e. fully deliverable. Please note, if a NRIS Network Upgrade entirely or partially changes the scope of an ERIIS Network Upgrade, only the cost difference between the NRIS upgrade and the ERIIS upgrade will be eligible for NRIS Network Upgrade cost allocation. Detailed NRIS Network Upgrade cost allocation calculations are provided in Section 9.

Table 8.1-2 – Network Upgrades Needed to Address MISO Identified NRIS Steady-State Injection Constraints

NRIS Thermal Constraints				Required NRIS Network Upgrades	Mitigation Type	Responsible Generators	NRIS Network Upgrade Cost Estimates (\$) [a]	ERIS Network Upgrade (Required for Same Constraint) Cost Estimate (\$) [b]	Previous Cycle ERIS/NRIS Network Upgrade (Identified for Same Constraint) Cost Estimate (\$) [c]	Cost Used for NRIS Network Upgrade Allocation (\$) [c] = [a] - [b]
699360 NLK GV T	138	699267 BRLGTN1	138 1	North Lake Geneva Tap - Burlington 138 kV (6541), Reconductor	NR	J1042, J1188	3,963,594	0	0	3,963,594

8.2 Per Project Summary

8.2.1 J986

J986 Deliverable (NRIS) Amount in 2023 Case: 149.76 MW (Conditional on ERIS upgrades and case assumptions)	149.76 MW Requested								
Next Constraint for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Single Deduction	Minimum Level of Service Attainable (MW) *This would be the amount of service to be granted if none of the projects contributing to this constraint, reduced service levels or withdrew	Shared Deduction	Maximum level of service Attainable (MW) *This would be the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared deduction	Max DF	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project	Total NRIS Cost of Upgrade	Notes
		149.76		149.76					

8.2.2 J1000

J1000 Deliverable (NRIS) Amount in 2023 Case: 50 MW (Conditional on ERIS upgrades and case assumptions)	50 MW Requested								
Next Constraint for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Single Deduction	Minimum Level of Service Attainable (MW) *This would be the amount of service to be granted if none of the projects contributing to this constraint, reduced service levels or withdrew	Shared Deduction	Maximum level of service Attainable (MW) *This would be the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared deduction	Max DF	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project	Total NRIS Cost of Upgrade	Notes
		50		50					

8.2.3 J1002

J1002 Deliverable (NRIS) Amount in 2023 Case: 99 MW (Conditional on ERIS upgrades and case assumptions)	99 MW Requested								
Next Constraint for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Single Deduction	Minimum Level of Service Attainable (MW) *This would be the amount of service to be granted if none of the projects contributing to this constraint, reduced service levels or withdrew	Shared Deduction	Maximum level of service Attainable (MW) *This would be the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared deduction	Max DF	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project	Total NRIS Cost of Upgrade	Notes
		99		99					

8.2.4 J1003

J1003 Deliverable (NRIS) Amount in 2023 Case: 50 MW (Conditional on ERIS upgrades and case assumptions)	50 MW Requested								

Next Constraint for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Single Deduction	Minimum Level of Service Attainable (MW) *This would be the amount of service to be granted if none of the projects contributing to this constraint, reduced service levels or withdrew	Shared Deduction	Maximum level of service Attainable (MW) *This would be the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared deduction	Max DF	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project	Total NRIS Cost of Upgrade	Notes
		50		50					

8.2.5 J1042

J1042 Deliverable (NRIS) Amount in 2023 Case: 142.97 MW (Conditional on ERIS upgrades and case assumptions)	180								
Next Constraint for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Single Deduction	Minimum Level of Service Attainable (MW) *This would be the amount of service to be granted if none of the projects contributing to this constraint, reduced service levels or withdrew	Shared Deduction	Maximum level of service Attainable (MW) *This would be the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared deduction	Max DF	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project	Total NRIS Cost of Upgrade	Notes
699360 NLK GV T 138 699267 BRLGTN1 138 1	57.03	142.97	53	147	0.343	J1042, J1188	\$3,720,312	\$3,963,594	
		180		180					

8.2.6 J1101

J1101 Deliverable (NRIS) Amount in 2023 Case: 20 MW (Conditional on ERIS upgrades and case assumptions)	20 MW Requested								
Next Constraint for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Single Deduction	Minimum Level of Service Attainable (MW) *This would be the amount of service to be granted if none of the projects contributing to this constraint, reduced service levels or withdrew	Shared Deduction	Maximum level of service Attainable (MW) *This would be the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared deduction	Max DF	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project	Total NRIS Cost of Upgrade	Notes
		20		20					

8.2.7 J1153

J1153 Deliverable (NRIS) Amount in 2023 Case: 150 MW (Conditional on ERIS upgrades and case assumptions)	150 MW Requested								

Next Constraint for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Single Deduction	Minimum Level of Service Attainable (MW) *This would be the amount of service to be granted if none of the projects contributing to this constraint, reduced service levels or withdrew	Shared Deduction	Maximum level of service Attainable (MW) *This would be the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared deduction	Max DF	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project	Total NRIS Cost of Upgrade	Notes
		150		150					

8.2.8 J1154

J1154 Deliverable (NRIS) Amount in 2023 Case: 75 MW (Conditional on ERIS upgrades and case assumptions)	75 MW Requested								
Next Constraint for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Single Deduction	Minimum Level of Service Attainable (MW) *This would be the amount of service to be granted if none of the projects contributing to this constraint, reduced service levels or withdrew	Shared Deduction	Maximum level of service Attainable (MW) *This would be the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared deduction	Max DF	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project	Total NRIS Cost of Upgrade	Notes
		75		75					

8.2.9 J1171

J1171 Deliverable (NRIS) Amount in 2023 Case: 100 MW (Conditional on ERIS upgrades and case assumptions)	100 MW Requested								
Next Constraint for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Single Deduction	Minimum Level of Service Attainable (MW) *This would be the amount of service to be granted if none of the projects contributing to this constraint, reduced service levels or withdrew	Shared Deduction	Maximum level of service Attainable (MW) *This would be the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared deduction	Max DF	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project	Total NRIS Cost of Upgrade	Notes
		100		100					

8.2.10 J1188

J1188 Deliverable (NRIS) Amount in 2023 Case: 0 MW (Conditional on ERIS upgrades and case assumptions)	50 MW Requested								
Next Constraint for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Single Deduction	Minimum Level of Service Attainable (MW) *This would be the amount of service to be granted if none of the projects contributing to this constraint, reduced service levels or withdrew	Shared Deduction	Maximum level of service Attainable (MW) *This would be the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared deduction	Max DF	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project	Total NRIS Cost of Upgrade	Notes
699360 NLK GV T 138 699267 BRLGTN1 138 1	208.34	0	14.72	35.28	0.0807	J1042, J1188	\$243,282	\$3,963,594	
		50		50					

8.3 Network Upgrade Alternatives Considered

All of the NRIS network upgrades identified in Table 8.1-2 are direct upgrades of the constraint facilities to ATC design standards and considered as least-cost solutions at this point. Therefore, no other alternatives were examined.

9.0 COST ALLOCATION

The cost allocation of Network Upgrades for the study group reflects responsibilities for mitigating system impacts based on Interconnection Customer-elected level of Network Resource Interconnection service as of the System Impact Study report date. The cost estimate for each network upgrade was provided by the corresponding transmission owning company.

9.1 ERIS Network Upgrades Proposed for Apr 18 DPP Phase 3 Projects

Network upgrades for Energy Resource Interconnection Service (ERIS) were identified in the ERIS analysis. The ERIS network upgrades include thermal network upgrades and voltage support network upgrades identified in the steady-state analysis, stability network upgrades identified in the dynamic stability analysis, short circuit network upgrades identified in the short circuit analysis. EMT stability network upgrades identified in the PSCAD analysis and network upgrades identified in the affected system analysis. For DPP April 2018 East (ATC) Phase 3 group, the total costs of network upgrades for the 2023 scenario are summarized in Tables 9.1-1, 9.1-2, 9.1-3, 9.1-4, 9.1-5, 9.1-6 and 9.1-7.

Table 9.1-1 – ERIS Network Upgrades Identified in Steady State Analysis

Steady-State Injection Constraint	Facility Owner	Network Upgrade	ISD (For Cost Estimate Only)	Cost (\$)¹,²
Sand Lake 138/69 kV transformer	ATC	Sand Lake 138/69 kV transformer, replacement	09/01/2020	3,105,696
Wautoma 138/69 kV transformer	ATC	Wautoma 138/69 kV transformer, replacement	09/01/2020	2,978,370
J986 POI – Port Edwards 138 kV line	ATC	J986 POI – Port Edwards 138 kV (X-11), uprate	09/01/2020	584,677
Saratoga 138/115 kV transformer T31	ATC	MTEP ProjID 18017: Petenwell-Badger West-Saratoga 138kV (X-43), Rebuild	12/31/2022	0
Jefferson – Crawfish River 138 kV line	ATC	Jefferson – Crawfish River 138 kV (6654), uprate	04/01/2021	957,298
Crawfish River – Concord 138 kV line	ATC	Crawfish River – Concord 138 kV (9043), uprate	04/01/2021	902,919

¹ All Network Upgrades were estimated on the earliest ISD dollars of responsible generator.

² ATC Network Upgrades included a 10% contingency.

Table 9.1-2 – ERIS Network Upgrades in Dynamic Stability Analysis

Constraint	Facility Owner	Network Upgrade	Cost (\$)
None	-	-	-

Table 9.1-3 – ERIS Network Upgrades in Short Circuit Analysis (Substations Under MISO Functional Control)

Substation	Facility Owner	Tenants for Joint Use Substations	Total Cost (\$)
HANCOCK	Alliant	ATC	220,000
HOLLYWOOD	Alliant	ATC	220,000
ROEDER	Alliant	ATC	220,000
SAND LAKE	ATC	-	220,000
WAUTOMA	ATC	Alliant	220,000
SHEEPSKIN	ATC	Alliant	220,000
NORTH LAKE GENEVA	ATC	Alliant	220,000
NORTH BEAVER DAM	ATC	Alliant	220,000
PORT EDWARDS	ATC	Alliant	220,000
SARATOGA	ATC	Alliant	220,000
CHAFFEE CREEK	ATC	-	220,000
HOWARDS GROVE	ATC	Alliant	220,000
PLYMOUTH SUB 4	Plymouth Utilities	ATC	220,000
HOLLAND	We Energies	ATC	220,000
BUTTERNUT	We Energies	ATC	220,000

Table 9.1-4 – Potential Affected System Substation Ground Grid Upgrades

Substation	Facility Owner	Total Cost (\$)
ACEC HANCOCK	Adams-Columbia Coop	To be determined by Affected System Facility Owner
ACEC WAUTOMA	Adams-Columbia Coop	
ACEC RICHFORD	Adams-Columbia Coop	
ACEC WILD ROSE PUMP	Adams-Columbia Coop	
PLAINFIELD (ALTE)	Alliant	
DANA CORP	Alliant	
SAUNDERS CREEK	Alliant	
SILVER LAKE (ALTE)	Alliant	
VULCAN CHEM	Alliant	
BEAVER DAM 3RD ST	Alliant	
CENTER ST	Alliant	
REC FULTON	Rock Energy Coop	

Table 9.1-5 – ERIS Network Upgrades Identified in PSCAD Analysis

Electromagnetic Transient Instability Identified for the following generators	Facility Owner	Network Upgrade	ISD (For Cost Estimate Only)	Cost (\$)
None	-	-	-	-

Table 9.1-6 – PJM Affected System Upgrades

Constraint	Facility Owner	Network Upgrade	Cost (\$)
None	-	-	-

Table 9.1-7 – ERIS Shared Network Upgrades

Constraint	Facility Owner	Network Upgrade	Cost (\$)
None	-	-	-

9.2 NRIS Network Upgrades Proposed for Apr 18 DPP Phase 3 Projects

Network upgrades for Network Resource Interconnection Service (NRIS) were identified in the MISO's deliverability analysis and listed in the Table 9.2-1 below.

Table 9.2-1 – NRIS Network Upgrades Identified

Network Upgrade	Facility Owner	ISD (For Cost Estimate Only) ¹	Cost Used for NRIS Cost Allocation (\$) ²
North Lake Geneva Tap - Burlington 138 kV (6541), Reconductor	ATC	8/31/2021	3,963,594

¹All Network Upgrades were estimated on the earliest ISD dollars of responsible generators.

²ATC Network Upgrades included a 10% contingency.

9.3 Cost Allocation Methodology for Thermal Network Upgrades

The costs of Network Upgrades (NU) for a set of generation projects (one or more subgroups or entire group with identified NU) are based off the MW impact of the worst-case scenario for each specific generator project. Basically, whatever the highest MW impact (increasing flow) is for that particular generator where the constraint is identified and requires NU is how it should be calculated.

Constraints which are mitigated by one or a subset of NU are identified. The highest MW contribution on these constraints from each generating facility is calculated in the MISO DPP study models without any Network Upgrades. Then the cost of each NU is allocated based on the pro rata share of the MW contribution from each generating facility on the constraints mitigated or partly mitigated by this NU. The methodology to determine the cost allocation of NU is:

$$\text{Project A cost portion of NU} = \text{Cost of NU} \times \left(\frac{\text{Max}(\text{Proj. A MW contribution on constraint})}{\sum_i \text{Max}(\text{Proj. i MW contribution on constraint})} \right)$$

9.4 Cost Estimating and Allocation Methodology for Short Circuit Upgrades

For each ATC breaker shown to be overdutied a new breaker will be scoped and the cost of that upgrade will be assigned to generators. No breaker duty analysis is performed for Affected System equipment. Appendix N provides short circuit data for all substation buses 69 kV and above included in ATC protection CAPE models that show a 1% increase in SLG or 3PG fault current. Any non-ATC equipment requiring a change would be considered Affected System.

For each ATC substation that shows a bus having fault current (SLG) increased by 10% or more, ATC will perform detailed grounding grid analysis if in-house grounding grid modeling is available at the time of the study. For the substations that pass the detailed grounding analysis (i.e. meet the safety criteria), they will be removed from the cost allocation process. For the remaining substations, costs are assigned for ground grid upgrades at that substation.

Only one cost estimate is scoped per substation regardless of the number of buses at that substation that show a 10% or greater increase in fault current. The largest fault current value at a substation in the After Case will determine the ground grid upgrade costs. If the largest fault current is above 20 kA, the upgrades are assigned a cost estimate of \$550,000. If the largest fault current is at or below 20 kA, the upgrades are assigned a cost estimate of \$220,000 for ground grid upgrades only. These are placeholder costs until Facility Studies are performed. Based on those results, costs may be adjusted.

The ground grid study methodology (10% or greater increase in SLG fault current) is applied to both ATC and non-ATC facilities modeled in ATC Protection models. For non-ATC facilities, it is interconnection customer's responsibility to work with facility owners to develop costs, and implementation of the mitigation projects (if needed).

Once breaker replacement costs and substation ground grid upgrade costs are determined, they are allocated proportionally to study generators that have a greater than 3% of the total of all current queue study generator contributing fault currents under the single line-ground short circuit fault simulation at the identified substation.

9.5 Cost Allocation Tables

The cost allocation of Network Upgrades for the study group reflects responsibilities for mitigating system impacts based on Interconnection Customer-elected level of Network Resource Interconnection service as of the issued date of the System Impact Study report.

Assuming all generating facilities in the DPP 2018 April East (ATC) Phase 3 group advance, Table 9.5-1 - Table 9.5-7 show how the costs for ERIIS steady-state Network Upgrades, ERIIS short-circuit Network Upgrades, potential Affected System substation ground grid upgrades, ERIIS EMT stability Network Upgrades, PJM Affected System upgrades and NRIS Network Upgrades allocated to responsible generating facilities.

Table 9.5-1 – ERIS Steady-State Thermal Network Upgrade Costs Allocated to Each Generation Project

Required ERIS Network Upgrades	Worst MW Impact, % Cost Allocation			Total MW Impact	Cost Allocation (\$)		
	J986	J1002	J1154		J986	J1002	J1154
Sand Lake 138/69 kV transformer, replacement	26.61, 100.00	-	-	26.61	3,105,696	-	-
Wautoma 138/69 kV transformer, replacement	29.00, 51.83	26.95, 48.17	-	55.95	1,543,689	1,434,681	-
J986 POI – Port Edwards 138 kV (X-11), uprate	106.16, 75.92	33.66, 24.08	-	139.82	443,887	140,790	-
MTEP PrjID 18017: Petenwell-Badger West-Saratoga 138kV (X-43), Rebuild	19.65, 100.00	-	-	19.65	0	-	-
Jefferson – Crawfish River 138 kV (6654), uprate	-	-	9.87, 100.00	9.87	-	-	957,298
Crawfish River – Concord 138 kV (9043), uprate	-	-	9.87, 100.00	9.87	-	-	902,919
Total ERIS Steady State Thermal Network Upgrade Cost (\$) Allocated to Each Generator					5,093,272	1,575,471	1,860,217

Table 9.5-2 – ERIS Short-Circuit Network Upgrade % Cost Allocation to Each Generation Project
(Substations Under MISO Functional Control)

SUBSTATION	Facility Owner	Tenants for Joint Use Substations	Total Cost (\$)	% Cost Allocation										
				J986	J1000	J1002	J1003	J1042	J1101	J1153	J1154	J1171	J1183	J1188
HANCOCK	Alliant	ATC	220,000	48.55%	-	51.45%	-	-	-	-	-	-	-	-
HOLLYWOOD	Alliant	ATC	220,000	94.11%	-	5.89%	-	-	-	-	-	-	-	-
ROEDER	Alliant	ATC	220,000	20.32%	-	73.34%	6.34%	-	-	-	-	-	-	-
SAND LAKE	ATC	-	220,000	51.45%	-	48.55%	-	-	-	-	-	-	-	-
WAUTOMA	ATC	Alliant	220,000	21.67%	-	78.33%	-	-	-	-	-	-	-	-
SHEEPSKIN	ATC	Alliant	220,000	-	-	-	-	-	-	-	-	-	-	100%
NORTH LAKE GENEVA	ATC	Alliant	220,000	-	-	-	-	100%	-	-	-	-	-	-
NORTH BEAVER DAM	ATC	Alliant	220,000	-	-	5.54%	94.46%	-	-	-	-	-	-	-
PORT EDWARDS	ATC	Alliant	220,000	100%	-	-	-	-	-	-	-	-	-	-
SARATOGA	ATC	Alliant	220,000	90.24%	-	9.76%	-	-	-	-	-	-	-	-
CHAFFEE CREEK	ATC	-	220,000	37.60%	-	62.40%	-	-	-	-	-	-	-	-
HOWARDS GROVE	ATC	Alliant	220,000	-	-	-	-	-	-	100%	-	-	-	-
PLYMOUTH SUB 4	Plymouth Utilities	ATC	220,000	-	-	-	-	-	-	100%	-	-	-	-
HOLLAND	We Energies	ATC	220,000	-	-	-	-	-	-	100%	-	-	-	-
BUTTERNUT	We Energies	ATC	220,000	-	-	-	-	-	-	-	-	100%	-	-

Table 9.5-3 – ERIS Short-Circuit Network Upgrade Costs Allocated to Each Generation Project
(Substations Under MISO Functional Control)

SUBSTATION	Facility Owner	Tenants for Joint Use Substations	Total Cost (\$)	Cost Allocation (\$)										
				J986	J1000	J1002	J1003	J1042	J1101	J1153	J1154	J1171	J1183	J1188
HANCOCK	Alliant	ATC	220,000	106,810	-	113,190	-	-	-	-	-	-	-	-
HOLLYWOOD	Alliant	ATC	220,000	207,042	-	12,958	-	-	-	-	-	-	-	-
ROEDER	Alliant	ATC	220,000	44,704	-	161,348	13,948	-	-	-	-	-	-	-
SAND LAKE	ATC	-	220,000	113,190	-	106,810	-	-	-	-	-	-	-	-
WAUTOMA	ATC	Alliant	220,000	47,674	-	172,326	-	-	-	-	-	-	-	-
SHEEPSKIN	ATC	Alliant	220,000	-	-	-	-	-	-	-	-	-	-	220,000
NORTH LAKE GENEVA	ATC	Alliant	220,000	-	-	-	-	220,000	-	-	-	-	-	-
NORTH BEAVER DAM	ATC	Alliant	220,000	-	-	12,188	207,812	-	-	-	-	-	-	-
PORT EDWARDS	ATC	Alliant	220,000	220,000	-	-	-	-	-	-	-	-	-	-
SARATOGA	ATC	Alliant	220,000	198,528	-	21,472	-	-	-	-	-	-	-	-
CHAFFEE CREEK	ATC	-	220,000	82,720	-	137,280	-	-	-	-	-	-	-	-
HOWARDS GROVE	ATC	Alliant	220,000	-	-	-	-	-	-	220,000	-	-	-	-
PLYMOUTH SUB 4	Plymouth Utilities	ATC	220,000	-	-	-	-	-	-	220,000	-	-	-	-
HOLLAND	We Energies	ATC	220,000	-	-	-	-	-	-	220,000	-	-	-	-
BUTTERNUT	We Energies	ATC	220,000	-	-	-	-	-	-	-	-	220,000	-	-
Total ERIS Short Circuit NU Cost (\$) Allocated to Each Generator				1,020,668	0	737,572	221,760	220,000	0	660,000	0	220,000	0	220,000

Table 9.5-4 – Potential Affected System Substation Ground Grid Upgrades Allocated to Each Generation Project

SUBSTATION	Facility Owner	Total Cost (\$)	% Cost Allocation										
			J986	J1000	J1002	J1003	J1042	J1101	J1153	J1154	J1171	J1183	J1188
ACEC HANCOCK	Adams-Columbia Coop	To be determined by Affected System Facility Owner	48.55%	-	51.45%	-	-	-	-	-	-	-	-
ACEC WAUTOMA	Adams-Columbia Coop		23.03%	-	76.97%	-	-	-	-	-	-	-	-
ACEC RICHFORD	Adams-Columbia Coop		31.19%	-	68.81%	-	-	-	-	-	-	-	-
ACEC WILD ROSE PUMP	Adams-Columbia Coop		23.13%	-	76.87%	-	-	-	-	-	-	-	-
PLAINFIELD (ALTE)	Alliant		51.42%	-	48.58%	-	-	-	-	-	-	-	-
DANA CORP	Alliant		-	-	-	-	-	-	-	-	-	-	100%
SAUNDERS CREEK	Alliant		-	-	-	-	-	-	-	-	-	-	100%
SILVER LAKE (ALTE)	Alliant		22.93%	-	77.07%	-	-	-	-	-	-	-	-
VULCAN CHEM	Alliant		100%	-	-	-	-	-	-	-	-	-	-
BEAVER DAM 3RD ST	Alliant		-	-	6.29%	93.71%	-	-	-	-	-	-	-
COLOMA	Alliant		43.08%	-	56.92%	-	-	-	-	-	-	-	-
CENTER ST	Alliant		-	-	6.34%	90.49%	-	-	-	3.17%	-	-	-
REC FULTON	Rock Energy Coop		-	-	-	-	-	-	-	-	-	-	100%

10.0 AVAILABLE APPENDIX DOCUMENTS (NOT ATTACHED)

Appendix A – Study Criteria, Methodology, and Assumptions

Appendix B – ATC Planning Criteria and Generation Facility Interconnection Guide

Appendix C – Interconnection Facility Project Diagrams

Appendix D – Network Upgrade Project Diagrams

Note: Project Diagrams were not developed for line upgrade projects.

Appendix E – Steady State Power Flow Results

Appendix F – Steady State Operating Restriction Study Results

Appendix G – Dynamic Stability Results

Appendix H – PJM Affected System Study Results

Appendix I – MISO Deliverability Study Results

Appendix J – Assessed System Performance Reference

Appendix L – PSCAD Study Results

Appendix M – Conditions to GIA

Appendix N - Short Circuit Data

